

The Practical Guide to Audio Engineering

From Vibrating Air to a Finished Mix

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Chapter 1 — What Sound Actually Is

Before we talk about kilohertz, codecs or cabinets, we need a shared picture of what sound *is*. Almost every concept in this book is a consequence of three ideas: sound is a wave, the wave has a **frequency** and an **amplitude**, and real instruments produce many frequencies at once. Get these straight and the rest of the book falls into place.

1.1 Sound is moving air

When a guitar string vibrates, it pushes the air next to it, which pushes the air next to *that*, and so on — a travelling pattern of tiny pressure changes that eventually reaches your eardrum and pushes it in and out. Your brain interprets that motion as sound.

A microphone does exactly what your eardrum does: it has a thin diaphragm that moves with the air pressure, and it turns that motion into a fluctuating electrical voltage. That voltage is the **analog audio signal** — a continuous electrical “copy” of the air-pressure wave. Everything downstream (cables, preamps, converters) is just handling that voltage.

***Key idea:** audio gear never touches “sound.” It handles an electrical representation of a pressure wave. Quality is about how faithfully each stage preserves or intentionally alters that representation.*

1.2 Frequency — how fast it wiggles (pitch)

Frequency is how many times the pressure cycles back and forth per second, measured in **hertz (Hz)**. One full compression-and-rarefaction per second is 1 Hz. A thousand cycles per second is one **kilohertz (kHz)**.

Frequency is what we perceive as **pitch**:

- Low frequency = low pitch (a bass note, a kick drum).
- High frequency = high pitch (a cymbal, a whistle).

Some anchors worth memorizing:

Frequency	What it is
20 Hz	The lowest pitch most humans can hear (felt more than heard)
~41 Hz	Low E on a 4-string bass
~82 Hz	Low E on a guitar (open 6th string)
440 Hz	“A above middle C” — the standard tuning reference (A4)
1,000 Hz (1 kHz)	The ear’s most sensitive region
~4 kHz	“Presence” / clarity / consonants in speech
20,000 Hz (20 kHz)	The top of human hearing (and it drops with age)

That **20 Hz – 20 kHz** range is the headline number: it’s the span of human hearing, and it’s the reason digital audio is built the way it is (Chapter 2).

Pitch is **logarithmic**, not linear. Every time you *double* the frequency, you go up one **octave**. 110 Hz, 220 Hz, 440 Hz, 880 Hz are all the note “A,” each an octave apart. This is why audio frequency charts and EQ displays are spaced logarithmically — equal *musical* steps take equal *visual* space.

1.3 Amplitude — how big it wiggles (loudness)

Amplitude is how far the pressure swings from rest — the *size* of the wave. Bigger swings move your eardrum more and you perceive them as **louder**.

Loudness, like pitch, is roughly logarithmic, so we measure it in **decibels (dB)** — a ratio scale, not an absolute one. A few practical facts:

- +6 dB $\approx$ “twice the voltage/amplitude.”
- +10 dB $\approx$ “about twice as loud” to human perception.
- The decibel always describes a *ratio* between two levels. In digital audio we measure relative to the maximum possible level and call it **dBFS** (decibels relative to **F**ull **S**cale), so digital levels are negative numbers counting *down* from 0 (the ceiling). More on that in Chapter 2.

The difference between the quietest and loudest sound a system can handle is its **dynamic range** — a concept that comes back when we discuss bit depth.

1.4 Why a piano and a guitar playing the same note sound different

Here’s the most important idea in the chapter. Play A2 (110 Hz) on a bass, a piano and a trumpet — same pitch, completely different sound. Why?

Because real instruments almost never produce a single, pure frequency. A pure single-frequency tone is a **sine wave**, and it sounds sterile and electronic (it’s the “test tone” beep). Instead, a vibrating string or air column produces a **fundamental** frequency *plus* a whole series of higher frequencies called **harmonics** (or **overtones**).

The harmonic series

The harmonics are (very nearly) integer multiples of the fundamental:

Harmonic	Frequency (for 110 Hz fundamental)	Musical relationship
1st (fundamental)	110 Hz	the note itself
2nd	220 Hz	one octave up
3rd	330 Hz	octave + a fifth
4th	440 Hz	two octaves up
5th	550 Hz	two octaves + a major third
...	...	...

The *pitch* you hear is set by the fundamental. But the *relative loudness of each harmonic* — which ones are strong, which are weak, how they fade over time — is what your brain reads as **timbre** (tone color). That recipe of harmonics is the instrument’s sonic fingerprint.

This single idea explains an enormous amount of what's later in the book:

- **EQ** (Chapter 9) works by boosting or cutting *bands* of frequencies — it reshapes the harmonic balance.
- **Distortion** in a preamp (Chapter 5) sounds “warm” or “harsh” precisely because it *adds new harmonics* that weren't in the original signal.
- A **guitar cabinet** (Chapter 6) sounds the way it does because it *removes* certain harmonics — a speaker can't reproduce the very high ones.

1.5 The shape of a sound over time: the envelope

Two sounds can have the same pitch and the same harmonics and still differ, because sound changes *over time*. The **envelope** describes how loudness evolves from the moment a note starts:

- **Attack** — how fast it reaches full volume (a plucked string is instant; a bowed violin swells in).
- **Decay / Sustain** — how it settles and holds.
- **Release** — how it fades after you let go.

A piano has a sharp attack and a long decay; an organ has a slow attack and a flat sustain. The envelope is why a piano note played backwards sounds nothing like a piano. Compression (Chapter 10) is, at heart, a tool for *reshaping the envelope*.

1.6 Phase — when waves combine

When two copies of a wave meet, they add together sample-by-sample. If their peaks line up (**in phase**), they reinforce and get louder. If one's peak meets the other's trough (**out of phase**), they cancel and get quieter — push it to a perfect mirror image and they cancel to *silence*.

You don't need the math, but you need the instinct, because phase quietly governs real situations:

- Two microphones on one guitar cabinet can partially cancel each other, thinning the tone, depending on their distance apart.
- A stereo mix collapsed to mono can lose elements that were relying on phase differences.
- Convolution and IRs (Chapter 7) are entirely an exercise in adding up delayed, phase-shifted copies of a signal.

Summary

- Sound is a travelling pressure wave; audio gear works on an electrical copy of it.
- **Frequency (Hz)** = pitch; human hearing spans ~**20 Hz to 20 kHz**; pitch is logarithmic (double the frequency = up one octave).
- **Amplitude** = loudness, measured in **decibels (dB)**, a ratio scale.
- Real instruments produce a **fundamental + harmonics**; the harmonic recipe is **timbre**, and reshaping it is what EQ and distortion do.
- The **envelope** (attack/decay/sustain/release) shapes a sound over time.

- **Phase** governs how multiple copies of a wave combine — reinforcing or cancelling.
-

***Try this.** Open any tone-generator app or your DAW's test-tone plugin. Play a 110 Hz sine — notice how thin and “electronic” it is. Now play a real bass note at the same pitch. The pitch matches, but the bass is rich and alive because of its harmonics. That difference is timbre, and it's the thing every later chapter is trying to capture, preserve, or shape.*

Chapter 2 — From Air to Numbers: Sample Rate & Bit Depth

In Chapter 1 we turned a pressure wave into a continuous electrical voltage. To get that into a computer we have to turn the smooth, continuous signal into a list of numbers. That conversion — done by the **A/D converter** (analog-to-digital) in your audio interface — is defined by two settings you'll choose every time you start a project: **sample rate** and **bit depth**. This chapter explains exactly what they mean, the famous **44.1 kHz**, and what to actually set them to.

2.1 Sampling: measuring the wave many times per second

A computer can't store a smooth curve; it stores numbers. So the converter *measures* the incoming voltage at regular intervals and writes down each value. Each measurement is a **sample**. The **sample rate** is how many measurements it takes per second, in hertz.

- **44.1 kHz** means the signal's amplitude is measured **44,100 times every second**.
- **48 kHz** means 48,000 times per second.

Play those samples back fast enough through a D/A converter (digital-to-analog, on the way out to your speakers) and your ear can't tell the staircase of values from the original smooth wave. This whole scheme is called **PCM** — Pulse Code Modulation — and it's the foundation of WAV files, CDs, and virtually all uncompressed digital audio (Chapter 3).

2.2 The Nyquist theorem — why 44.1 kHz and not less

How many samples per second do you actually *need*? The answer is one of the cornerstones of digital audio: the **Nyquist–Shannon sampling theorem** (1940s).

*To perfectly capture a signal, you must sample at **more than twice** the highest frequency present in that signal.*

The highest frequency a given sample rate can represent is half the sample rate, called the **Nyquist frequency**.

Now connect it to Chapter 1: human hearing tops out around **20 kHz**. To capture everything we can hear, we need a sample rate just above $2 \times 20 \text{ kHz} = 40 \text{ kHz}$. That's why **44.1 kHz** exists — 40 kHz of audio plus a little extra room above 20 kHz for the anti-aliasing filter to do its work. At 44.1 kHz the Nyquist frequency is **22.05 kHz**, comfortably above the limit of hearing.

(The oddly specific **.1** is a historical artifact of how early digital audio was stored on video tape, but the practical takeaway is simple: 44.1 kHz captures the full audible band.)

2.3 Aliasing — what happens when you don’t sample fast enough

If a frequency in the signal is *higher* than the Nyquist frequency, the converter can’t represent it correctly. Instead it gets misread as a *different, lower* frequency — a false “alias” that wasn’t in the original sound. This is **aliasing**, and it’s the digital equivalent of wagon wheels appearing to spin backwards in old films.

Aliasing sounds bad — inharmonic, metallic, ugly — because the false frequencies have no musical relationship to the real ones. To prevent it, converters put an **anti-aliasing filter** (a steep low-pass filter) right before sampling, removing anything above the Nyquist frequency so it can never fold back. This is also why you should think about sample rate inside plugins that generate harmonics (distortion, saturation): they can create frequencies above Nyquist and alias, which is why such plugins often **oversample** internally.

2.4 Bit depth — how precisely each sample is measured

Sample rate is *how often* we measure. **Bit depth** is *how precisely* we record each measurement — how many distinct amplitude values are available to describe the loudness of each sample.

- **16-bit** gives $2^{16} = 65,536$ possible levels per sample.
- **24-bit** gives $2^{24} = 16,777,216$ possible levels per sample.
- **32-bit float** uses a different scheme (a floating-point number) that gives effectively gigantic range — more on that below.

Think of bit depth as the number of “rungs” on the ladder the converter uses to describe amplitude. More rungs = a finer, more accurate description of quiet detail. Too few rungs and the smooth wave gets forced onto coarse steps, introducing **quantization error** — heard as a gritty distortion on quiet passages.

Bit depth = dynamic range, not “warmth”

The thing bit depth actually controls is **dynamic range**: the distance between the loudest possible sample (0 dBFS, the ceiling) and the noise floor.

- 16-bit $\approx$ **96 dB** of dynamic range.
- 24-bit $\approx$ **144 dB** of dynamic range (in theory; in practice limited by analog noise, but well beyond any room or converter).

A useful rule of thumb: **each bit adds ~6 dB of dynamic range**. More bit depth does *not* make the sound “higher resolution” in pitch or add detail you can hear directly — it lowers the noise floor and gives you room to record at safe, conservative levels without losing quiet detail.

2.5 dBFS, headroom and why you record quieter than you think

Digital level is measured in **dBFS** — decibels below Full Scale. **0 dBFS is the absolute ceiling**. Go above it and samples get **clipped** (flattened at the maximum), producing harsh digital distortion that is *not* the musical kind from Chapter 5 — it’s just broken.

So when recording you leave **headroom**: aim peaks around **-12 to -18 dBFS**. This sounds quiet, and beginners instinctively push levels near 0 “to be loud,” but:

- Loudness is set later, in mixing/mastering — not at the converter.
- 24-bit gives you so much dynamic range that recording at -18 dBFS still leaves the noise floor inaudibly low.
- Headroom protects against an unexpectedly loud peak clipping and ruining a take.

This is the single most important practical takeaway of the chapter: **record with healthy headroom**. 24-bit is what makes that free.

2.6 32-bit float: the “can’t clip” format

Many modern interfaces and field recorders now offer **32-bit float**. Because of how floating-point numbers work, the usable dynamic range is enormous — so large that, in practice, you effectively *cannot clip* a 32-bit float recording. If a peak goes “over 0,” the information is still there in the file and you can simply pull the level back down afterwards with no damage. It’s a genuine convenience for tracking unpredictable sources, at the cost of larger files. For DAW *processing*, audio is internally handled in 32-bit (or 64-bit) float anyway, which is why turning a plugin’s gain down after a loud one doesn’t destroy the signal.

2.7 Dither — the friendly noise

When you *reduce* bit depth (e.g., exporting a 24-bit mix to a 16-bit file for CD/streaming), the quantization error can become an audible, correlated distortion on fade-outs and quiet tails. **Dither** is a tiny amount of carefully shaped random noise added during that bit-depth reduction. It “randomizes” the error, trading nasty distortion for a far less noticeable, constant low-level hiss. Rule: **apply dither only once, at the final export to a lower bit depth**. Don’t dither at every step.

2.8 So what should I actually set?

Scenario	Sample rate	Bit depth
Music-only project (most home studios)	44.1 kHz	24-bit
Audio for video / film / broadcast	48 kHz	24-bit
Tracking unpredictable live sources	44.1 or 48 kHz	32-bit float (if available)
Final export for streaming/CD	44.1 kHz	16-bit (with dither)
“Hi-res” / heavy pitch-shifting work	88.2 / 96 kHz	24-bit

Practical guidance:

- **24-bit is non-negotiable for recording.** The extra headroom it buys is the real benefit, and storage is cheap.
- **44.1 kHz is plenty for music.** Use **48 kHz** if your project will sync to video (it's the video-world standard).
- **Higher sample rates (88.2/96 kHz) have real but modest benefits** — mainly more room above the audible band for distortion/oversampling plugins and extreme time-stretching. They double your file sizes and CPU load. Most great records were and are made at 44.1/48 kHz.
- **Pick one sample rate per project and stay there.** Mixing files of different rates forces sample-rate conversion, which is best avoided.

Summary

- Digital audio = measuring the wave's amplitude many times per second (**sample rate**) at some precision (**bit depth**). This is **PCM**.
- **Nyquist:** the highest capturable frequency is half the sample rate. Human hearing ends near 20 kHz, so **44.1 kHz** (Nyquist 22.05 kHz) captures it all.
- Sampling too slowly causes **aliasing** — false lower frequencies; an **anti-aliasing filter** prevents it.
- **Bit depth = dynamic range** (~6 dB per bit). 16-bit ≈ 96 dB, 24-bit ≈ 144 dB.
- Stay below **0 dBFS**; record with **headroom** around -12 to -18 dBFS.
- **32-bit float** is effectively unclippable; **dither** is added only at the final reduction to a lower bit depth.
- Defaults: **44.1 kHz / 24-bit** for music, **48 kHz / 24-bit** for video.

***Try this.** In your DAW, record the same phrase twice: once peaking near 0 dBFS and once peaking around -18 dBFS. Then normalize both to the same loudness and compare. On a 24-bit project they'll sound essentially identical — proof that you gain nothing by recording hot, and everything by keeping headroom.*

Chapter 3 — Audio Formats & Compression

You now know that digital audio is a stream of PCM numbers (Chapter 2). A raw stream of those numbers is large — about **10 MB per minute** for CD-quality stereo. **File formats** are the different ways we package and shrink that data. The big divide is between formats that keep *every* number (**lossless**) and formats that throw some away to save space (**lossy**). This chapter explains how each works and which to use when.

3.1 The two kinds of compression

The word “compression” is overloaded in audio. There are two completely different meanings:

1. **Data compression** (this chapter): making the *file* smaller. Splits into **lossless** (perfectly reversible) and **lossy** (permanently discards data).
2. **Dynamic-range compression** (Chapter 10): an *effect* that reduces the volume difference between loud and quiet parts of a sound.

They are unrelated. Here we mean file size.

Within data compression:

- **Uncompressed:** every PCM sample stored verbatim (WAV, AIFF).
- **Lossless compression:** repacked smaller but bit-for-bit identical when decoded — like ZIP for audio (FLAC, ALAC). No quality loss.
- **Lossy compression:** permanently deletes audio data the ear is least likely to notice, for dramatically smaller files (MP3, AAC, Ogg Vorbis, Opus). Once gone, it can’t be recovered.

3.2 Uncompressed: WAV and AIFF

Both **WAV** (Microsoft/IBM) and **AIFF** (Apple) are containers for raw PCM. They are functionally equivalent in quality — they store exactly the samples your converter produced, no more, no less.

- **Quality:** perfect (it *is* the original data).
- **Size:** large. ~10 MB/min stereo at 44.1 kHz/16-bit; more at 24-bit/higher rates.
- **Use them for:** recording, editing, mixing, and archiving masters. This is your **working format** — never edit or master from a lossy file.
- WAV is the universal interchange format; AIFF is its Apple-flavored twin. A practical caveat: standard WAV files have a 4 GB size limit (fine for songs, a problem for very long recordings), where formats like RF64/BWF extend it.

3.3 Lossless compression: FLAC and ALAC

These shrink the file *without losing anything* — decode them and you get the exact original PCM back. They do it by encoding the data more cleverly (predicting samples and storing only the differences, then entropy-coding the result), much like a ZIP archive.

- **FLAC** (Free Lossless Audio Codec): open, royalty-free, widely supported. Typically compresses to ~50–60% of the WAV size. The standard for lossless distribution and archiving.
- **ALAC** (Apple Lossless): Apple’s equivalent, for the Apple/iTunes ecosystem. Similar performance.
- **Quality:** perfect — identical to WAV on decode.
- **Size:** roughly half of WAV.
- **Use them for:** archiving libraries, distributing high-quality downloads, and high-resolution streaming tiers where you want full fidelity at a smaller size.

3.4 Lossy compression: MP3, AAC, Ogg Vorbis, Opus

This is where most everyday listening happens. Lossy codecs exploit **psychoacoustics** — the science of what the ear can and can’t perceive — to decide what to delete:

- **Frequency masking:** a loud tone hides a nearby quieter tone, so the quiet one can be discarded.
- **Temporal masking:** a loud sound hides quiet sounds immediately before and after it.
- **Hearing limits:** content near/above 20 kHz can be reduced or removed.

The encoder spends its limited data budget on what you *can* hear and throws away the rest. How big that budget is, is set by the **bitrate**.

Bitrate: the quality dial for lossy files

Bitrate = how many kilobits of data per second (**kbps**) the file uses. Higher bitrate = more data kept = better quality and bigger files.

Bitrate (MP3/AAC)	Rough quality
128 kbps	Acceptable; artifacts audible to trained ears, esp. on cymbals/reverb
192 kbps	Good for casual listening
256 kbps	Very good; the iTunes/AAC standard
320 kbps	Highest standard MP3; near-transparent for most listeners

Bitrate can be **constant (CBR)** or **variable (VBR)**. VBR spends more bits on complex passages and fewer on simple ones, giving better quality per megabyte — prefer VBR when the option exists.

The codecs

- **MP3** — the original mass-market format. Universally compatible; plays *everywhere*. At 320 kbps it's hard to distinguish from lossless in casual listening. Slightly dated, but unbeatable for compatibility.
- **AAC** (Advanced Audio Coding) — MP3's successor; **better quality than MP3 at the same bitrate**. The default for Apple Music, YouTube, and most modern streaming/video. Files often carry the `.m4a` extension.
- **Ogg Vorbis** — open, patent-free, quality competitive with AAC. Famously used by Spotify (up to 320 kbps) and in games.
- **Opus** — the modern open standard; **the best low-to-mid-bitrate quality available**, excellent for both music and voice, and the default for many modern streaming, voice-chat and web (WebRTC) applications. If you control both ends, Opus is technically the strongest choice.

The golden rule of lossy: never re-encode

Each lossy encode discards data. Encoding an MP3 *from* another MP3 (“transcoding”) compounds the damage — artifacts stack. Always make lossy files **once, from your lossless master**. Keep WAV/FLAC as the source of truth.

3.5 Which format for which job?

Job	Best choice	Why
Recording / editing / mixing	WAV (or AIFF)	Lossless, universal, the working master
Archiving your library/masters	FLAC	Lossless, ~half the size of WAV
High-quality download / hi-res streaming	FLAC / ALAC	Full fidelity, smaller
Everyday portable listening	AAC 256 kbps or MP3 320 kbps	Tiny files, transparent enough
Maximum compatibility (old gear, any player)	MP3	Plays on literally everything
Voice, podcasts, low-bandwidth streaming	Opus	Best quality per kbps
Sending a quick demo for feedback	MP3 256–320 kbps	Small, emails/uploads fast
Delivering a final master to a client/label	WAV 24-bit	They need the full-quality source

3.6 A note on streaming and “loudness normalization”

Streaming platforms (Spotify, Apple Music, YouTube) deliver lossy streams (commonly AAC or Ogg/Opus at various tiers) and additionally apply **loudness normalization** — they turn everything to a common loudness target (measured in **LUFS**) on playback. The practical consequence for you as a producer: making a master *brick-wall loud* gains you nothing on streaming (it gets turned down anyway) and costs you dynamics. Master to the platform’s target (commonly around **-14 LUFS**), and let the dynamics breathe.

Summary

- A file format is how PCM audio is packaged; the key split is **lossless** vs. **lossy**.
- **WAV/AIFF** = uncompressed, perfect, large — your **working/master** format.
- **FLAC/ALAC** = lossless compression, ~half size, perfect quality — archiving and hi-fi distribution.
- **MP3/AAC/Ogg/Opus** = lossy, much smaller, quality set by **bitrate**; they use **psychoacoustic masking** to discard inaudible data. AAC beats MP3 at equal bitrate; Opus is best per kilobit; MP3 wins on compatibility.
- **Never master from, or re-encode, a lossy file.** Keep lossless as the source.
- For streaming, master to a **LUFS target**, not maximum loudness.

***Try this.** Take one WAV file and export it as MP3 at 128, 192 and 320 kbps, plus a FLAC. Compare file sizes, then A/B them on good headphones — focus on cymbals, vocal sibilance and reverb tails, where lossy artifacts hide. Notice how hard 320 kbps is to tell from the FLAC, and how 128 kbps “smears” the highs. That’s psychoacoustic compression at work.*

Chapter 4 — Anatomy of a Guitar Tone

People say “I want a Marshall tone” or “that’s a Fender sound,” as if tone comes from one box. It doesn’t. The sound that reaches a listener is the result of a **chain** of stages, each one shaping the signal — and each one is responsible for a recognizable part of the final tone. This short chapter maps the whole chain so the next four chapters (preamps, cabinets, IRs, NAM) have a place to hang. Even if you play bass or keys, this is the model for thinking about *any* amplified or processed signal.

4.1 The chain, end to end

Here is the classic electric-guitar signal path, in order:

Guitar	→	Pedals	→	Preamp	→	Power amp	→	Cabinet	→	Microphone	→	
Audio interface												
(strings & pickups)		(drive, EQ, FX)		(gain & tone-shaping)		(push & "feel")		(speaker + box, a filter)		(captures the moving air)		(A/D → DAW)

Each arrow is a transformation. Let’s walk it.

1. The instrument: strings and pickups

The string vibrates (Chapter 1: fundamental + harmonics), and magnetic **pickups** convert that vibration into a weak electrical signal. Pickup type (single-coil vs. humbucker), position (neck = warm, bridge = bright), and even the wood and the player’s hands set the *raw material* every later stage works on. Garbage in, garbage out: tone starts at the fingers.

2. Pedals / effects

Optional stomp-boxes between guitar and amp: **overdrive/distortion** (add harmonics and sustain), **EQ/boost**, **wah**, **compressor**, and time/ modulation effects. Where an effect sits in the chain matters — we cover the “why” of each effect type in Chapters 10–11.

3. The preamp

The first amplification stage. It boosts the tiny pickup signal and, crucially, **shapes the tone and adds the gain/distortion character** you associate with a particular amp. This is where “the sound of the amp” largely lives. **Chapter 5 is entirely about why the preamp changes the sound.**

4. The power amp

Takes the preamp’s line-level signal and boosts it to the high power needed to physically move a speaker. At high volumes the power amp adds its own compression and “sag” and feel — part of why a cranked amp feels alive. (In modeling, power-amp behavior is often folded into the captured model.)

5. The cabinet (speaker + box)

The speaker turns electricity back into moving air — but it’s a *terrible, beautiful* filter. A guitar speaker can’t reproduce the full 20 Hz–20 kHz range; it rolls off the harsh highs and shapes the mids, and the wooden box adds its own resonance. **The cabinet is responsible for a huge part of the final tone** — Chapter 6.

6. The microphone (and the room)

A mic in front of the cabinet captures the moving air. Mic *choice* and *placement* dramatically change what’s recorded — moving a mic an inch changes the tone more than most EQ moves. The room adds reflections and ambience.

7. The audio interface

The mic’s analog voltage hits the interface’s **mic preamp** (different from the guitar amp’s preamp — this one just brings the level up cleanly) and then the **A/D converter** (Chapter 2), which turns it into the digital samples your DAW records.

4.2 The digital revolution: collapsing the chain

Everything above can be done with physical gear in a room. But two technologies let you reproduce most of that chain *in the box*, and they map directly onto the stages above — which is why they get their own chapters:

- **Impulse Responses (IR)** capture the **cabinet + mic + room** stage — the *linear* part of the chain — as a small file you can apply with a convolution plugin. (Chapter 7)
- **Neural Amp Modeler (NAM)** captures the **preamp + power amp** stage — the *nonlinear*, distortion-generating part — using a neural network. (Chapter 8)

Put a NAM model (the amp) in front of an IR (the cab) and you’ve rebuilt the whole “amp → cab → mic” chain digitally, from a clean DI’d guitar signal. This amp-then-cab order is the key to understanding both technologies, and it’s why we split them into “nonlinear” (NAM) and “linear” (IR) — a distinction we’ll define precisely in Chapters 7 and 8.

4.3 Why “where it happens” matters

Two reasons this map is worth internalizing:

1. **Diagnosing tone problems.** “It’s too harsh” — is that the pickup, the preamp gain, the cab/IR, or the mic position? Knowing the chain tells you where to reach. Fizzy high-end is usually a cab/IR or mic issue, not a preamp-gain issue, for instance.
2. **Order is non-commutative.** Running distortion *into* an EQ sounds different from EQ *into* distortion, because distortion adds harmonics based on whatever it’s fed. The same logic explains why amp-before-cab is the only order that makes physical sense.

Summary

- Guitar tone is a **chain**: instrument → pedals → preamp → power amp → cabinet → mic → interface. Each stage owns a recognizable slice of the sound.
- The **preamp** contributes most of the amp's gain/tone character (Chapter 5); the **cabinet+mic** shapes and tames the result (Chapter 6).
- Digitally, **NAM** captures the nonlinear **amp** stage (Chapter 8) and **IR** captures the linear **cab+mic** stage (Chapter 7); chained amp → cab, they rebuild the whole rig in the box.
- Signal-chain **order matters**, because each stage transforms what the next one receives.

***Try this.** Record a clean DI (direct) guitar signal once. Then audition it through (a) just a NAM amp model with no cab, (b) just an IR cab with no amp, and (c) amp → cab together. Hearing each stage in isolation makes the chain concrete: the amp brings the drive and feel, the cab brings the “in the room” filtering, and only together do they sound like a record.*

Chapter 5 — Why Preamps Change the Sound

The preamp is the first amplifier stage in the chain (Chapter 4), and it's where most of "the sound of the amp" is born. But *why* does a preamp do more than make the signal louder? Why does a tube preamp sound "warm" and a cranked one "crunchy"? The answer is **gain staging** and **harmonic distortion** — and once you understand them, distortion pedals, overdrive, saturation plugins and even analog "warmth" all click into place at once.

5.1 The preamp's two jobs

A preamp does two things:

1. **Amplification.** Your pickups output a tiny signal (millivolts). The preamp boosts it up toward line level so the rest of the chain has something to work with.
2. **Tone shaping.** This is the interesting part. The way a preamp amplifies is not perfectly clean — and those imperfections are the tone. The preamp also houses the **tone stack** (the bass/mid/treble EQ controls).

If a preamp amplified *perfectly* — every output an exact scaled-up copy of the input — it would be "transparent" and add no character. Real preamps don't, and that's the whole point.

5.2 Linear vs. nonlinear: the core idea

This single distinction explains everything in this chapter (and sets up Chapters 7–8):

- A **linear** process changes a signal's *level* or *frequency balance* but **never creates new frequencies** that weren't already there. A volume knob and an EQ are linear.
- A **nonlinear** process **creates new frequencies** — specifically, new **harmonics** — that were not in the input.

A preamp pushed hard is **nonlinear**. That's why it doesn't just get louder; it changes *timbre*, because it's adding harmonics to the signal (recall from Chapter 1 that the harmonic recipe *is* timbre).

5.3 Clipping: where the new harmonics come from

Every amplifier has a maximum output swing it can produce — a ceiling and a floor set by its power supply. Feed it a signal small enough and it amplifies cleanly. Push the input harder and the peaks of the wave try to go beyond what the amp can deliver, so the tops get **flattened, or "clipped."**

Here's the crucial fact from the math of sound: **flattening the peaks of a wave is exactly the same thing as adding harmonics to it.** A pure sine wave that gets its tops squared off is, mathematically, the original sine *plus* a series of new harmonic frequencies. The more you flatten it, the more (and stronger) the added harmonics — the more "distorted" it sounds.

So:

- **A little clipping** = a few gentle added harmonics = “warmth,” “drive,” “grit.”
- **A lot of clipping** = many strong added harmonics = “crunch,” “fuzz,” full “distortion.”

That’s the entire mechanism. Overdrive, distortion, fuzz, “saturation” and tube “warmth” are all the same phenomenon — clipping that adds harmonics — differing only in *how* the clipping happens.

5.4 The shape of the clipping is the sound

Not all clipping sounds alike, because *how* the wave is flattened decides *which* harmonics get added and how strongly. Two factors dominate:

Hard vs. soft clipping

- **Soft clipping** rounds the peaks off gradually. It adds harmonics that fall away smoothly — musical, smooth, “tube-like” overdrive.
- **Hard clipping** squares the peaks abruptly. It adds lots of strong high-order harmonics — aggressive, buzzy, “solid-state” or heavy-distortion character.

Even vs. odd harmonics — the famous “tube warmth”

This is the heart of the tube-vs-solid-state debate.

- **Even-order harmonics** (2nd, 4th, ...) are *octaves and octave-related* intervals above the fundamental. The **2nd harmonic is exactly one octave up** — maximally consonant. Even harmonics sound **warm, full, musical, pleasant**.
- **Odd-order harmonics** (3rd, 5th, ...) add a more dissonant, “edgy” or “hollow” character. A little is aggressive and exciting; a lot is harsh.

Tube (valve) circuits tend to clip **asymmetrically** — they round one half of the wave differently from the other — which generates a strong dose of **even** harmonics, especially that octave-up 2nd harmonic. That’s the mechanism behind why tube amps are described as “warm” and “musical”: they’re literally adding the most consonant possible overtone. Even a tube amp’s “*clean*” sound contains a small amount of this harmonic distortion (often a few percent THD), which is part of why “clean tube tone” still sounds richer than a truly transparent signal.

Solid-state (transistor) circuits historically clipped more symmetrically and abruptly, favoring **odd** harmonics and a harder edge — the root of the old “solid-state sounds harsh” reputation. (Modern designs and good modeling can voice solid-state circuits to be smooth too — it’s about the clipping behavior, not magic.)

***Takeaway:** “tube warmth” isn’t mysticism. It’s predominantly **even-order harmonic distortion** from **asymmetric, soft clipping** — and the octave-up 2nd harmonic is doing most of the pleasing work.*

5.5 Gain staging: cascading the drive

Modern high-gain amps don’t get their massive distortion from one stage slammed hard. They use **multiple gain stages in series** — the output of one feeding the input of the next — with level controls between them. This is **gain staging** or **cascading gain**.

Why cascade instead of one big stage? Because feeding an already-clipped, harmonically-rich signal into *another* clipping stage produces a smoother, more complex, more controllable distortion than one brutal stage would. Each stage adds and reshapes harmonics. It also lets the designer place EQ (tone shaping) *between* stages, so the drive can be tightened (e.g., trimming low end before a high-gain stage to avoid a flubby, muddy mess).

The trade-off: every gain stage also amplifies **noise** and hiss. Sloppy gain staging = a noisy, fizzy tone; good gain staging = tight, harmonically rich, and relatively quiet. This is exactly why “gain” and “master volume” are separate controls — gain sets how hard you drive the *preamp* (how much distortion), master sets the overall *output* level.

5.6 Bias, dynamics and “feel”

A couple of finer points that explain why real preamps feel responsive:

- **Bias** sets the operating point of a tube. “Cold” biasing pushes the tube toward earlier, crunchier, harder clipping; “hot/warm” biasing gives a smoother response. It’s another lever on the clipping shape, hence the tone.
- **Touch sensitivity.** Because clipping depends on how hard the stage is driven, a preamp distorts *more* when you pick hard and *less* when you play softly or roll back your guitar’s volume. This dynamic interaction — the amp “cleaning up” when you back off — is a huge part of what players mean by “feel,” and it’s exactly the nonlinear, input-dependent behavior that NAM (Chapter 8) works so hard to capture and that a static EQ never could.

5.7 Why this matters for recording in the box

Because preamp distortion is **nonlinear and dynamic**, you cannot fake it with EQ (which is linear) or a simple volume change. That’s the entire reason **Neural Amp Modeler** exists: a neural network is needed to learn the complex, input-level-dependent way a real preamp adds harmonics. An IR (linear convolution) captures the cab perfectly but *cannot* capture preamp distortion — which is precisely why the two technologies are separate and complementary (Chapters 7–8).

Summary

- A preamp **amplifies** and, when driven, **shapes tone by adding harmonics** — a **nonlinear** process (it creates frequencies that weren't there).
- The mechanism is **clipping**: flattening a wave's peaks is mathematically identical to adding harmonics. More clipping = more/stronger harmonics = more distortion.
- The *shape* of clipping decides the sound: **soft** clipping = smooth/musical, **hard** = aggressive; **even** harmonics (esp. the octave-up 2nd) = "warmth," **odd** = "edge."
- **Tube warmth** $\approx$ even-order harmonics from asymmetric soft clipping.
- **Gain staging** (cascaded stages with EQ between them) builds smooth, controllable high-gain tones — at the cost of added noise.
- The distortion is **dynamic** (responds to playing intensity), which is why it's "feel," and why it needs neural modeling, not EQ, to reproduce.

***Try this.** Put a clean amp model on a guitar DI and slowly raise the gain while keeping the output level matched. Listen for the moment it stops just getting louder and starts getting richer/grittier — that's the onset of clipping adding harmonics. Then play a chord hard, then soft: hear it distort more on hard hits. You're hearing nonlinearity and touch-sensitivity directly.*

Chapter 6 — Why Cabinets Change the Sound

If the preamp (Chapter 5) is the heart of the amp’s *character*, the cabinet is the final, decisive shaper of the tone that actually reaches a microphone. Players underestimate it constantly — but plug the same amp into two different cabs, or move the mic two inches, and the change dwarfs almost any EQ move. Here’s why the cabinet is, by some estimates, *half your tone*.

6.1 A guitar speaker is a (wonderful) bad speaker

A hi-fi speaker is engineered to reproduce the full audible range (Chapter 1: 20 Hz–20 kHz) as flatly and accurately as possible. A **guitar speaker is the opposite** — it’s deliberately limited and colored, and that’s the point.

A typical guitar speaker:

- **Rolls off the lows** below roughly **80–100 Hz** — it physically can’t move enough air down there, which keeps the tone from being a muddy rumble.
- **Rolls off the highs** sharply above roughly **5–6 kHz**. This is the single most important thing a cab does. Remember from Chapter 5 that distortion adds lots of *high-order* harmonics — much of that energy lands as harsh, fizzy “buzz” up high. The speaker’s high-frequency roll-off **filters that fizz away**, turning a raw, buzzy distortion into a smooth, musical tone.
- **Has bumps and dips through the mids** — resonant peaks (often a presence bump somewhere around 1–5 kHz and a body resonance down low) that give a speaker its recognizable “voice.”

So the cabinet is essentially a **strong, characterful filter** (Chapter 9’s EQ is the same idea, applied deliberately). This is exactly why an unmic’d, “amp sim with no cab” tone sounds unbearably harsh and fizzy — there’s nothing filtering the high-order distortion harmonics. **The cab tames the amp.**

6.2 The box matters too

A speaker doesn’t hang in mid-air; it’s mounted in a **cabinet**, and the box shapes the sound further:

- **Open-back** cabs (common in combo amps) let sound radiate from the rear, giving a more open, airy, spacious tone with less low-end punch.
- **Closed-back** cabs (common 4×12 stacks) trap the rear wave, producing tighter, punchier, more focused low end — the classic rock/metal “thump.”
- **Size, wood, and number of speakers** add resonances and interactions. Four speakers in a 4×12 couple together for a bigger low end than a single 1×12.

This is **linear** behavior (Chapter 5’s distinction): the cabinet changes the *frequency balance* and adds resonances, but it doesn’t create brand-new harmonics the way a clipping preamp does. That linearity is *exactly* what makes the cabinet capturable as an Impulse Response — hold that thought for Chapter 7.

6.3 The microphone is part of the cabinet sound

When recording, you almost never hear “the cab” directly — you hear **the cab through a microphone**. And the mic, plus where you put it, is a massive tone control:

Mic choice

Different mics have different frequency responses and characters. The legendary **Shure SM57** dynamic mic is the studio default on guitar cabs — its built-in presence bump flatters electric guitar. A ribbon mic sounds smoother and darker; a condenser, more detailed and airy. The mic imposes its own “EQ curve” on what’s captured.

Mic placement — the biggest free tone knob you have

Moving a single mic around the speaker changes the tone more than most EQ:

- **Center of the cone (on-axis, “dead center”)** = brightest, most aggressive, most high-end and “bite.”
- **Toward the edge of the cone / off-axis** = progressively warmer, darker, rounder, less fizz.
- **Distance from the grille** = closer is tighter with more proximity bass; pulling back adds room sound and air, and changes the low-end via the **proximity effect**.
- **Angle** relative to the speaker further tilts the high-frequency content.

A speaker’s tone isn’t uniform across its surface — the center radiates more highs than the edge — so the mic is sampling a *specific spot* of a *specific speaker*. That’s why “move the mic an inch” is real advice, not folklore.

The room

Finally, the room the cab sits in adds reflections, ambience and its own resonances. A close mic captures mostly the cab; a room mic captures the interaction of cab + space.

6.4 Putting it together: the cab is the linear half of the chain

Step back and look at the chain from Chapter 4. The tone-defining work splits cleanly into two halves:

Stage	Behavior	What it does	How it’s modeled digitally
Preamp / power amp	Nonlinear	Adds harmonics (distortion, “warmth,” feel)	NAM (Chapter 8)
Cabinet + speaker + mic + room	Linear	Filters/shapes frequencies, adds resonance, removes fizz	IR (Chapter 7)

Because the cabinet+mic+room is a **linear filter**, its *entire* effect can be captured as a single short recording — an **Impulse Response** — and re-applied to any signal perfectly. That’s the bridge to the next chapter, and the reason IRs are so small, so CPU-cheap, and so faithful.

Summary

- A guitar speaker is **deliberately band-limited**: it rolls off lows (~below 80–100 Hz) and, crucially, **highs (~above 5–6 kHz)**, which **removes the harsh fizz** of distortion’s high-order harmonics. It’s a strong, characterful filter.
- The **cabinet box** (open vs. closed back, size, materials, speaker count) adds resonance and shapes the low end.
- You record the cab **through a mic**, so **mic choice and placement** (center vs. edge, distance, angle) are enormous tone controls — often bigger than EQ.
- All of this is **linear** behavior (frequency shaping, not new harmonics) — which is exactly why a cabinet+mic can be captured as an **Impulse Response** (Chapter 7), the linear counterpart to NAM’s nonlinear amp capture.

***Try this.** Load one IR loader on a distorted guitar and switch between an “on-axis/center” capture and an “off-axis/edge” capture of the same cabinet. Then try a 1×12 vs. a 4×12 IR. With the amp untouched, the tone transforms — proof that the cab and mic position are doing at least as much work as the amp.*

Chapter 7 — Impulse Responses (IR)

In Chapter 6 we established that the cabinet + speaker + mic + room is a **linear filter** — it shapes frequencies but doesn’t create new harmonics. This chapter explains the technology that captures that filter perfectly in a tiny file: the **Impulse Response (IR)**, applied through **convolution**. IRs are how you get a real, mic’d-cabinet sound from a digital amp, and they’re one of the most useful and CPU-cheap tools in modern production.

7.1 What an IR is

An **Impulse Response** is a recording of how a system responds to a single, instantaneous burst of sound — an “impulse.” Think of it as an **acoustic fingerprint** or a photograph of a space/device’s sonic character.

The idea: if you could feed a *perfect* impulse (an infinitely short, full- spectrum click containing every frequency at once) into a linear system and record what comes out the other side, that recording would contain **everything about how that system colors sound** — every frequency boost and cut, every resonance, every reflection. That output recording *is* the impulse response.

For a guitar cabinet, the IR captures the whole linear chain at once:

the speaker’s frequency response + the cabinet’s resonance + the microphone’s character + the mic placement + (optionally) the room.

A typical cabinet IR is only a **few milliseconds long** and a few kilobytes in size, yet it contains all of that information.

7.2 Why this works: linearity (and why amps don’t qualify)

The reason an impulse response *completely* describes a system rests on one property from Chapter 5: **linearity**.

A linear system has a magic property — its behavior is **fully described by how it responds to a single impulse**. Once you know the impulse response, you can predict the system’s output for *any* input whatsoever. Speaker cabinets, rooms, and many time-based effects are (approximately) linear, so a single IR captures them entirely.

This is also precisely **why an IR cannot capture a preamp/amp’s distortion**. A distorting amp is **nonlinear** — its response *depends on how hard you hit it* (Chapter 5’s touch sensitivity), so there’s no single fixed “response” to capture. Hence the division of labor:

Captured by	System	Behavior
IR (this chapter)	Cabinet, mic, room, reverb	Linear
NAM (next chapter)	Preamp, power amp, distortion	Nonlinear

7.3 Convolution: applying the fingerprint

How do you *use* an IR? Through a mathematical operation called **convolution**, performed by a **convolution plugin** or **IR loader**.

Conceptually, convolution **“stamps” the IR’s character onto your dry signal**. The IR loader takes your incoming (dry, un-cabbed) signal and the IR file, and combines them so the output sounds exactly as if your signal had physically passed through the original cabinet + mic + room.

Mechanically (you don’t need the math, but the intuition is useful): convolution slides the IR along your signal, and for every sample of input it adds in a tiny, scaled, delayed copy of the entire IR, summing them all up. Because the IR encodes the system’s complete frequency-and-time response, this sum reproduces exactly what the real system would have output. Every sample of your guitar gets “painted” with the cabinet’s fingerprint.

Two happy consequences of the cabinet being linear:

- **IR files are tiny** (milliseconds) and **convolution is very CPU-efficient** — far cheaper than physically modeling a speaker.
- The result is **extremely faithful** — for a linear system, convolution with its IR is essentially *perfect* reproduction, not an approximation.

7.4 How IRs are captured

You can make your own, and understanding how demystifies them. There are two common methods:

1. **Sine sweep method (most common, highest quality)**. Play a smooth tone that sweeps from low to high frequency (“a sweep”) through the mic’d cabinet and record the result. Special software then mathematically *deconvolves* the recording against the known sweep to extract the impulse response. Sweeps give a cleaner, higher signal-to-noise IR than a raw impulse.
2. **Direct impulse method**. Feed an actual short burst/click through the system and record it. Simpler, but noisier and less precise — rarely used for commercial cab IRs.

Either way, the captured IR bakes in *every* choice you made: that specific speaker, that mic, that exact placement, that room. Want a different mic position? Capture a new IR. This is why IR libraries ship dozens of variations of one cabinet (SM57 on-axis, ribbon off-axis, blended pairs, etc.).

7.5 Using an IR loader in practice

The workflow in any DAW:

1. Get a **dry amp signal** — either a re-amped DI through a real amp head with no cab, or (more commonly today) a **NAM amp model with the cab/IR turned off** (Chapter 8). The golden rule: **the IR replaces the cabinet, so the signal feeding it must not already contain a cab.**
2. Insert an **IR loader / convolution plugin** (e.g., the free *NadIR*, *Cab*, *LeCab*, the loader built into many amp sims, or a generic convolution reverb) after the amp.
3. **Load an IR file** (`.wav` is the usual container; some loaders use proprietary formats).
4. **Audition and blend.** Try different IRs; many loaders let you load two IRs and blend them (mimicking two-mic setups), adjust phase/distance, and high-/low-pass the result to taste.

Order in the chain: **amp/distortion first, then IR** — exactly mirroring the physical reality of amp → cabinet (Chapter 4). Putting an IR *before* distortion would be like distorting the *speaker's* output, which isn't how a real rig works.

7.6 IRs aren't just for cabs

Because *any* linear system has an impulse response, the same technology powers:

- **Convolution reverb** — IRs of real concert halls, churches, plates and rooms, letting you place a sound in a real acoustic space.
- **Acoustic instrument body IRs** — capturing the resonant body of an acoustic guitar so a piezo-pickup signal sounds like a mic'd guitar.
- **Hardware/outboard “fingerprints”** of EQs and other linear processors.

Same principle every time: capture a linear system's impulse response, convolve it onto your signal.

Summary

- An **IR** is a recording of a **linear** system's response to an impulse — an acoustic “fingerprint” of a cabinet + speaker + mic + room, only milliseconds long.
 - It works because **linear systems are fully described by their impulse response** — so one IR can predict the output for any input. (Nonlinear amps can't be captured this way — that's NAM's job.)
 - **Convolution** applies the IR to your dry signal, “stamping” the cabinet's character onto it — faithfully and very CPU-efficiently.
 - IRs are captured via **sine sweeps** (or direct impulses); each one bakes in a specific mic and placement.
 - Use an IR **after** the amp/distortion, fed a **cab-less** signal; the same tech also powers **convolution reverb** and acoustic-body simulation.
-

Try this. On a distorted DI guitar, first listen with **no cab** — harsh, fizzy, unusable (that's the unfiltered high-order distortion from Chapter 5). Now insert an IR loader and load any cab IR. The fizz vanishes and it sounds like a record. You just heard the linear cabinet filter do its job — and you did it with a few-kilobyte file.

Chapter 8 — Neural Amp Modeler (NAM) & Profiling

We’ve now built the two halves of digital tone. Chapter 7 captured the **linear** cabinet with an IR. This chapter captures the **nonlinear** amp — the preamp and power amp whose distortion and “feel” (Chapter 5) an IR fundamentally cannot reproduce. The tool is **Neural Amp Modeler (NAM)**, a free, open-source project that uses deep learning to clone real amps with state-of-the-art accuracy. It’s arguably the biggest shift in guitar recording of the last several years.

8.1 The problem NAM solves

Recall the linear/nonlinear split (Chapters 5–7):

- A cabinet is **linear** → fully captured by an IR.
- An amp’s distortion is **nonlinear** → its response *changes with how hard you play*, so there is no single fixed “response” to convolve. An IR can’t do it.

For decades, the only way to reproduce an amp in software was **circuit modeling (physical/component modeling)**: engineers study the actual electronic circuit and write equations that simulate every tube, resistor and capacitor. This works, but it’s labor-intensive, model-specific, and only as good as the engineers’ approximations of a messy analog reality.

NAM takes a completely different, almost cheeky approach.

8.2 How NAM works: learning by listening

Instead of modeling the *circuit*, NAM **models the behavior**. It treats the amp as a black box and asks a neural network to *learn* the relationship between what goes in and what comes out:

1. Play a known **input signal** (a clean DI guitar / test signal) into the real amp.
2. **Record the amp’s output** (the “wet” signal) simultaneously.
3. Train a **neural network** to transform the input into the output — i.e., to produce the same output the real amp did, for that input.
4. After training on a wide range of input dynamics, the network has learned the amp’s **complete nonlinear behavior** — its gain staging, its EQ curve, *and* crucially its **touch sensitivity** (how it distorts more on hard hits and cleans up on soft ones).

The trained result is a small file — a **NAM capture** (also called a NAM profile, model, or `.nam` file) — that, loaded into the free NAM plugin, behaves like the original amp, dynamics and all. The makers describe the result as “virtually indistinguishable from the real thing.”

Because it learns *behavior from audio* rather than approximating a circuit, NAM can capture not just amps but **pedals, preamps, and even entire signal chains** — anything you can play a signal through and record.

***The key distinction.** Traditional sims try to recreate what gear should sound like from its circuit. NAM **listens to the real gear and trains a network to behave exactly like it.** It's data-driven cloning, not physics simulation.*

8.3 NAM vs. IR vs. “profiling” — keeping terms straight

These get muddled constantly:

Term	Captures	Method	Linear?
IR	Cabinet + mic + room	Convolution of an impulse response	Linear
NAM	Amp / preamp / pedal / chain	Neural network trained on input → output audio	Nonlinear
Kemper / “Profiling”	Amp (commercial hardware)	Proprietary profiling (a different, older commercial take on the same goal)	Nonlinear

NAM and the older commercial “profilers” (like Kemper) share the *goal* — capture a real amp by analyzing its sound — but NAM is **free, open-source, runs as a plugin in your DAW, and uses modern neural networks**, which is a large part of why it spread so fast.

A complete digital rig is therefore: **NAM (the amp) → IR (the cab)**. NAM provides the nonlinear distortion and feel; the IR provides the linear cab/mic filtering (Chapter 7). Many NAM captures are made *amp-only* (taken from the amp’s DI/line out before any speaker), so you add your own IR afterward — giving you free choice of cabinet. Some captures are “full rig” (amp + cab baked in), in which case you don’t add an IR.

8.4 Making your own NAM capture

You can profile your own gear for free. The two common methods:

1. **Sine-sweep / reamp method.** A standardized input file is played through your gear and the output recorded, then uploaded for training.
2. **Dry/Wet (DI) method.** Split your guitar with a DI box and record the dry signal *and* the processed (amped) signal simultaneously, so the trainer has a perfectly aligned input/output pair.

You then **train** the model. Training runs a machine-learning optimization, which needs a GPU. You have options:

- **Train locally** with the open-source NAM trainer (needs a capable GPU).

- **Train in the cloud.** Services such as **TONE3000** train your captures on cloud GPUs (e.g., RTX 4090s) with no coding or local setup — you upload your recordings and download a finished `.nam` file. (Per their docs.)

The output is an open `.nam` file compatible with the free NAM plugin in any DAW.

What “training” means here. Profiling an amp is genuinely a small machine-learning task: you’re fitting a neural network’s parameters so its output matches recorded target audio — the same shape as any supervised-learning problem. The “model” you download is those learned weights.

8.5 The NAM ecosystem

- **The plugin** is free and open-source, available in standard formats (VST3, AU, etc.) for all major DAWs, plus standalone. Load a `.nam` capture and play.
- **Captures are abundant and mostly free.** Communities and libraries — **TONE3000** (thousands of captures and IRs), plus Discord/forum/Facebook groups — host models of everything from a Fender Reverb to a Marshall JCM800, Vox AC30, Roland JC-120 and countless boutique amps and pedals. (Counts and catalogs grow constantly.)
- **Hardware support.** Because the format is open, NAM models now run on some hardware floor units and modelers too, not just in the DAW.

8.6 Practical tips for using NAM

- **Match the input level.** NAM captures are nonlinear, so they care about how hard you drive them. Most NAM plugins show input/output level controls and a target; set your DI level so it matches what the capture was trained at, or the gain/feel will be off. (This is the digital echo of Chapter 5’s point that distortion depends on input level.)
- **Amp-only capture → add an IR.** If the model is amp-only, insert an IR loader after it (Chapter 7). If it’s a full-rig capture, don’t.
- **Feed it a clean DI.** NAM expects a dry guitar signal, just like a real amp’s input. Don’t feed it an already-distorted or already-cabbed signal.
- **CPU.** Bigger/“hi-fi” capture architectures sound marginally better but cost more CPU than “standard”/“lite” ones; pick to taste and track count.

Summary

- NAM captures the **nonlinear** amp (preamp/power amp/pedal/chain) that an IR cannot — including distortion and **touch-sensitive feel**.
- It works by **training a neural network on input → output audio** from real gear, modeling *behavior* rather than simulating the *circuit*.

- A complete in-the-box rig is **NAM (amp)** → **IR (cab)**; amp-only captures let you choose your own IR, full-rig captures bake the cab in.
 - You can **make your own** captures (sweep or dry/wet DI methods) and **train** them locally or via cloud GPU services like **TONE3000**; the output is an open `.nam` file for the free plugin.
 - The ecosystem is **free, open-source, and huge**, with thousands of community captures of classic and boutique gear.
-

***Try this.** Download the free NAM plugin and a couple of free captures (e.g., a clean Fender-style and a high-gain Marshall-style). On one DI guitar, A/B them, then add an IR after each. Pay attention to dynamics: dig in hard, then play softly and roll your guitar volume back — hear the capture distort and then clean up. That responsiveness is the nonlinear behavior the network learned, and it's exactly what an IR alone could never give you.*

Chapter 9 — EQ: Carving the Frequency Spectrum

Equalization (**EQ**) is the most-used tool in mixing, and it follows directly from Chapter 1: every sound is a blend of frequencies, and EQ lets you turn bands of those frequencies **up or down**. That's it. With it you remove problems (mud, harshness, rumble), help instruments fit together, and shape character. This chapter covers how EQ works and gives concrete starting points for **guitar, bass, keyboard and vocals** — but treat every number as a place to start, not a rule.

9.1 What EQ is (and what it isn't)

An EQ is a set of **filters** that boost or cut the level of chosen frequency ranges. It is a **linear** process (Chapter 5): it changes the *balance* of frequencies that already exist — it never adds new harmonics. (When a plugin *does* add harmonics, that's saturation/distortion, Chapter 10, not EQ.)

The mental model: picture the frequency spectrum (20 Hz on the left → 20 kHz on the right, spaced logarithmically per Chapter 1). EQ lets you push any region of that line up (boost) or down (cut).

9.2 The filter types

Every EQ is built from a handful of filter shapes:

- **High-pass filter (HPF) / low-cut.** Removes everything *below* a chosen frequency. The single most useful move in mixing — cleans out rumble and sub-mud. (Confusingly named: it *passes the highs*, i.e. cuts the lows.)
- **Low-pass filter (LPF) / high-cut.** Removes everything *above* a chosen frequency. Tames fizz, harshness, hiss.
- **Bell / peak.** Boosts or cuts a band centered on a frequency. The workhorse. Its width is set by **Q** (see below).
- **Shelf (high or low).** Boosts or cuts everything above (high shelf) or below (low shelf) a frequency by a fixed amount — like a gentle tilt. Great for broad “more air” or “more warmth” moves.
- **Notch.** A very narrow, deep cut to surgically kill a single problem frequency (a resonant ring, hum, feedback).

Q — bandwidth

Q sets how wide or narrow a bell filter is. **Low Q = wide** (affects a broad range, sounds natural and musical — good for tone shaping). **High Q = narrow** (affects a pinpoint range — good for surgically removing a specific problem). Rule of thumb: **boost wide, cut narrow**. Broad boosts sound musical; narrow cuts remove problems without thinning everything.

9.3 Subtractive vs. additive EQ

Two philosophies, both valid, but lead with the first:

- **Subtractive EQ (cut first).** Remove what's *wrong* — mud, boxiness, harshness, rumble. Cutting is less likely to cause harshness or clipping than boosting, and it cleans up a mix without adding energy. **Default to cutting.**
- **Additive EQ (boost to enhance).** Add what's *missing* — a touch of “air,” some presence, a little warmth. Use sparingly and broadly.

The “sweep and destroy” technique

To *find* a problem frequency: take a bell, set a **narrow, high-Q boost**, sweep it across the spectrum until the nasty resonance jumps out and sounds worst, then **flip that boost into a cut** and widen it slightly. This finds mud, boxiness, and harshness fast.

9.4 EQ in context: it's about the whole mix

The most important principle: **EQ for the mix, not for the soloed track.** An instrument that sounds amazing alone often clashes when everything plays together, because instruments **share frequency ranges** and fight (“masking” — Chapter 3's psychoacoustics in action). The art is **carving complementary spaces**: cut a little where one instrument should sit so another can own that range. Classic example: cut the bass guitar slightly where the kick drum lives, or carve a dip in the guitars where the lead vocal sits.

9.5 The frequency map (memorize these zones)

The whole spectrum, in plain language:

Zone	Range	Character	Common move
Sub-bass	20–60 Hz	Felt more than heard; rumble	HPF most non-bass tracks here
Bass	60–250 Hz	Fundamental weight, power	Where mud also lives
Low-mids	250–500 Hz	Warmth — or “ mud ”/boxiness”	Most common cut zone
Mids	500 Hz–2 kHz	Body, “honk,” attack of most instruments	Cut “boxy” ~400–600 Hz; “nasal” ~1 kHz
High-mids	2–6 kHz	Presence, clarity, attack, “bite”	Boost for cut-through; harshness lives here too
Highs	6–12 kHz	Brilliance, sibilance, detail	Sibilance ~5–9 kHz
Air	12–20 kHz	Openness, “expensive” sheen	Gentle high shelf for sparkle

Anchor to remember: **mud lives in the low-mids (~200–500 Hz); presence and clarity live in the high-mids (~2–6 kHz); harshness also lives in the high-mids, so the same region gives both magic and pain.**

9.6 Starting points by instrument

*Reminder: these are **starting points**, not rules. Sweep around every value and judge in the full mix. See **Appendix A** for a consolidated chart.*

Electric guitar

- **High-pass ~80–100 Hz.** Electric guitar has little useful content below this; cutting it clears room for bass and kick.
- **The power is in 200 Hz–1 kHz.** This midrange is where the “meat” and body of guitars live — don’t get it chasing scooped tones in a dense mix.
- **Mud / boxiness: cut around 300–500 Hz** if it sounds congested.
- **Presence / bite: boost gently around 2.5–5 kHz** to cut through.
- **Fizz: low-pass or shelf down above ~6–8 kHz** if the distortion is harsh (often a cab/IR issue — Chapter 6).
- For **multiple guitars**, carve them differently (one slightly brighter, one warmer) so they don’t mask each other.

Acoustic guitar

- **High-pass ~70–100 Hz** — mics pick up low rumble/handling that muddies the mix; clean it out.
- **Boom / mud: cut around 150–300 Hz** if it’s boomy (often from a tight recording space or proximity).
- **Boxiness: cut around 500 Hz–1 kHz** if it sounds “honky.”
- **Sparkle / string detail: gentle boost ~8–12 kHz** for shimmer.
- **Pick attack / clarity: ~2–5 kHz** if it needs to cut through.

Bass (guitar)

- **Fundamental and weight: 60–250 Hz.** The “magic” of bass lives roughly **80–200 Hz** — the fundamentals and the punch.
- **Sub / bottom: 40–80 Hz** for the deep extension (mind your speakers/room when judging this).
- **Boom: cut around 180–250 Hz** if it’s too “boomy,” while preserving the fundamentals below.
- **Definition / “growl” / finger or pick attack: 700 Hz–2.5 kHz.** This is what lets bass be *heard* on small speakers (phones, laptops) that can’t reproduce the lows.
- **String/pick clack: ~2.5–5 kHz** if you want articulation to poke through.
- Sit it with the **kick drum**: decide who owns the deep lows and carve the other slightly there so they don’t fight.

Keyboard / piano / synth

- **Most attention in 50–250 Hz** — treat boom and mud here to keep it lush and warm without clouding the mix.

- **High-pass** anything that doesn't need deep lows (e.g., a bright synth pad, organ) so it doesn't crowd bass and kick.
- **Mud / boxiness: cut around 200–500 Hz** if it's congesting the low-mids.
- **Presence / clarity: gentle boost ~3–6 kHz** to bring a piano or lead synth forward.
- **Air / sparkle: high shelf ~10–14 kHz** for a polished, open top end.
- Pianos and pads are wide-band and easily mask vocals/guitars — **carve a dip where the lead element sits** rather than turning the whole thing down.

Vocals

- **Always high-pass** — **minimum ~80–100 Hz** (higher for thin/female voices, lower for deep voices). Removes rumble, plosive thumps and proximity build-up.
- **Mud / “chestiness”: cut around 200–400 Hz** if it sounds thick or boxy.
- **Boxiness / honk: cut around 500 Hz–1 kHz** if needed.
- **Presence / intelligibility: boost ~3–6 kHz** so consonants and lyrics cut through (for many male voices, clarity sits ~4–9 kHz).
- **Warmth/body** for a voice often lives ~85–250 Hz (e.g. ~100–150 Hz for fullness) — boost gently if thin, but watch for mud.
- **Air / “expensive” sheen: high shelf ~10–16 kHz.**
- **Sibilance (“sss”/“shh”) lives ~5–9 kHz** — but don't fix it with EQ (you'd dull the whole voice); use a **de-esser** (Chapter 10) which only ducks the harsh frequency *when* the “s” happens.

9.7 Common EQ mistakes

- **Soloing too much.** EQ in context (9.4).
- **Boosting everything.** If you boost every track, you've changed nothing relative — and you've eaten your headroom. **Cut more than you boost.**
- **Over-narrow boosts.** Narrow boosts sound unnatural and ringy. Boost wide.
- **Ignoring the low end.** Un-HPF'd tracks pile up sub-mud you can't hear on small speakers but that wastes headroom. HPF generously.
- **EQ'ing the symptom, not the source.** Harsh fizzy guitar? Try a different IR/mic position (Chapter 6) before reaching for a steep cut.

Summary

- EQ boosts/cuts **bands of frequencies** using filters: **HPF/LPF, bell, shelf, notch**; **Q** sets bandwidth (**boost wide, cut narrow**).
 - **Cut first (subtractive)** to remove mud/harshness; **boost sparingly** to enhance. Use **sweep-and-destroy** to find problems.
 - **EQ in the full mix**, carving complementary spaces so instruments don't mask each other.
 - Know the map: **mud ~200–500 Hz, presence/clarity ~2–6 kHz, air >10 kHz.**
 - **High-pass almost everything** that isn't bass/kick (vocals/guitars ~80–100 Hz). Per-instrument starting points above and in **Appendix A**.
-

Try this. On any track, insert an EQ, make a narrow +6 dB bell, and sweep it slowly from lows to highs. Listen to how each region sounds — find the “mud” around 300 Hz, the “honk” around 1 kHz, the “harsh” around 3 kHz, the “air” up top. This single exercise builds the frequency intuition that every mixing decision relies on.

Chapter 10 — Effects: The Mixing Toolkit

EQ (Chapter 9) shapes *frequency*. The rest of the mixing toolkit shapes **dynamics, time and space, and harmonics**. This chapter explains each major effect family — what it does, the controls that matter, and when to reach for it — so that Chapter 11’s per-instrument recipes make sense. Knowing *why* an effect does what it does beats memorizing presets.

10.1 Dynamics: controlling loud and quiet

Compression

A **compressor** automatically **turns down the loud parts** of a signal, reducing the gap between loud and quiet (the **dynamic range**). After it “squashes” the peaks, you turn the whole thing back up — so the *quiet* parts end up louder relative to the peaks. The result: a more even, controlled, “present,” “glued” sound.

Why it matters: a raw vocal might swing wildly in level (some words loud, some buried). Compression evens that out so every word sits in the mix.

The controls:

- **Threshold** — the level above which compression kicks in. Lower threshold = more of the signal gets compressed.
- **Ratio** — how *hard* it turns down what’s above the threshold. 2:1 is gentle; 4:1 is moderate; 8:1+ is heavy; ∞ :1 is **limiting** (a brick wall).
- **Attack** — how *fast* it clamps down after a peak crosses the threshold. **Fast attack** catches transients (tames the pick/pluck/“snap”); **slow attack** lets the transient *through* first (keeps punch/attack) then compresses the body.
- **Release** — how fast it stops compressing after the signal drops back. Too fast = “pumping”/distortion; too slow = it never recovers.
- **Makeup gain** — turns the now-quieter signal back up to match.
- **Knee** — how gradually the ratio engages around the threshold (soft knee = smoother/transparent; hard knee = more obvious).

The attack/release interaction is really an **envelope** tool (Chapter 1): fast attack softens the attack of a note; slow attack emphasizes it. That’s how compression adds “punch” or “smoothness” depending on settings.

Watch for **over-compression**: squash too hard and you kill the life/dynamics, cause pumping, and raise the noise floor. A few dB of gain reduction is often plenty.

Limiter

A **limiter** is a compressor with a very high ratio and fast attack — it stops a signal from ever exceeding a ceiling. Used on the **master bus** to maximize loudness safely and prevent clipping (mind the LUFS point from Chapter 3 — don’t over-limit for streaming).

Gate / Expander

A **noise gate** is the opposite of a compressor at the bottom end: it **silences (or attenuates) a signal below a threshold**. Essential for high-gain guitar (Chapter 5's cascaded gain is *noisy* — a gate kills the hiss/hum between riffs) and for tightening drums or cutting bleed/mic spill.

De-esser

A **de-esser** is a specialized, frequency-targeted compressor that ducks only the **sibilance** range (~5–9 kHz, Chapter 9) and only *when* an “s”/“sh” hits — fixing harsh vocals without dulling the whole voice the way a static EQ cut would.

10.2 Time-based: depth and space

These create the sense of an instrument existing in a *place*.

Reverb

Reverb simulates the countless reflections a sound makes in a physical space — a room, hall, chamber, or metal plate. It adds **depth, size and ambience**, and “glues” elements into a shared space. (Convolution reverb, Chapter 7, uses IRs of real spaces; algorithmic reverb synthesizes the reflections.)

Key controls:

- **Type** — room (small, natural), hall (big, lush), plate (smooth, classic on vocals), spring (that surfy guitar-amp sound), chamber.
- **Decay / RT60** — how long the tail lasts. Short = intimate; long = epic.
- **Pre-delay** — the gap before the reverb starts. Longer pre-delay keeps the dry sound clear and *in front* of its own reverb (helps vocals stay intelligible).
- **Mix / wet-dry** — how much reverb. On a channel insert keep it low; better, use a **send** (see 10.5).
- **Size, diffusion, damping** — fine-tune the space and how bright the tail is.

Pro move (from the research): **EQ your reverb**. High-pass the reverb return (~200–300 Hz) so it doesn't add mud, and roll off some highs to make it sit warmer behind the dry signal.

Delay (echo)

Delay repeats the signal one or more times after a set interval. Uses range from obvious rhythmic echoes to subtle thickening.

- **Time** — usually synced to tempo (1/4, 1/8, dotted-1/8 — the classic “U2”/rhythmic guitar delay). Set in ms or note values.
- **Feedback** — how many repeats (how much output is fed back in). High feedback = many, trailing echoes.
- **Mix** — level of the repeats.
- **Slapback** — a single, very short (~80–120 ms) echo for rockabilly/vocal thickening.

Delay vs. reverb: **delay adds “wetness” and width without the wash of reverb**, keeping things clearer — pros often combine a little of each. And the **Haas effect**: a copy delayed under ~40 ms isn’t heard as a separate echo but adds **thickness and stereo width** — the basis of many widening tricks.

10.3 Modulation: movement and width

Modulation effects use a **low-frequency oscillator (LFO)** to *vary* something over time (pitch, timing, or phase), creating motion, thickness and width.

- **Chorus.** Duplicates the signal, slightly **detunes and delays** the copies, and modulates them — creating the impression of several instruments playing together. Adds **width and shimmer** (lush on clean guitar, synths, backing vocals).
- **Flanger.** A short, modulated delay mixed with the original creates a sweeping “jet plane” comb-filter whoosh.
- **Phaser.** Sweeps a series of phase shifts for a softer, swirling sweep (classic on funk guitar, electric piano).
- **Tremolo.** Modulates *volume* up and down (that surf/vintage pulsing).
- **Vibrato.** Modulates *pitch* up and down.

Modulation is mostly about **flavor and width**. A common trick: a touch of chorus on a doubled part to widen the stereo image.

10.4 Harmonic effects: saturation and distortion

Unlike everything above (mostly linear or time-based), these are **nonlinear** (Chapter 5): they **add harmonics**.

- **Saturation / tape / tube emulation.** Gentle harmonic distortion that adds warmth, density, “analog” character and perceived loudness without obvious distortion. Great glue on buses, bass, and vocals.
- **Distortion / overdrive / fuzz.** The bigger-dose version — the guitar-tone mechanism of Chapter 5, now used as a mix color (e.g., grit on a synth or parallel-distorted drums).
- **Exciter / enhancer.** Generates new high-frequency harmonics to add “air” and sparkle beyond what EQ-boosting existing content can do.

10.5 How effects are applied: inserts vs. sends

This routing choice (you’ll set it up in Chapter 12) is as important as the effects themselves:

- **Insert.** The effect sits *in* the channel; 100% of the signal passes through it. Use for things that should process the *whole* signal: EQ, compression, gates, distortion, de-essing.

- **Send / aux / bus.** The channel sends a *copy* to a separate FX track holding the effect (set the FX to 100% wet). Use for **reverb and delay** so that:
 1. **Multiple instruments share one reverb** → they sound like they're in the *same room* (cohesion), and it saves CPU.
 2. You can **EQ and compress the effect return** independently (10.2's reverb tips, and "ducking" the reverb under the vocal so lyrics stay clear).

Parallel processing uses the same send idea: blend a heavily-processed copy (e.g., crushed parallel compression, "New York" drums) under the natural dry signal for power *and* dynamics.

Signal-chain order (on a channel insert)

A common, sensible order: **Gate** → **EQ (subtractive)** → **Compression** → **EQ (tone/additive)** → **Saturation** → **Modulation** → **(sends to Delay/Reverb)**. Order isn't sacred, but remember Chapter 5: anything nonlinear (compression, saturation) reacts to what feeds it, so cleaning up *before* it usually helps.

Summary

- **Dynamics: compression** evens loud/quiet (threshold/ratio/attack/release /makeup — attack & release shape the *envelope*); **limiters** cap peaks; **gates** silence below a threshold (kills high-gain noise); **de-essers** tame sibilance.
- **Time-based: reverb** = space/depth (type, decay, pre-delay; *EQ your reverb!*); **delay** = echoes/width (time, feedback; slapback; Haas widening).
- **Modulation** (chorus, flanger, phaser, tremolo, vibrato) uses an LFO for **movement and width**.
- **Saturation/distortion/exciters** are **nonlinear** — they **add harmonics** for warmth, density and air.
- Apply whole-signal processors as **inserts**; apply reverb/delay via **sends** so instruments share a space and the effect can be EQ'd/compressed. **Parallel processing** blends a processed copy under the dry.

Try this. On a vocal, set up a reverb on a **send** (not an insert). First hear it with the reverb full-range — muddy. Now high-pass the reverb return at 250 Hz and roll off above 8 kHz. The vocal suddenly sits in a clean, warm space without clouding the mix. That's the difference between using an effect and mixing with one.

Chapter 11 — Effects Recipes by Instrument & Vocals

This is the “what do I actually put on it?” chapter. For each instrument it gives a **signal chain** (in order) plus **starting settings**, pulling together the EQ map (Chapter 9) and the effects toolkit (Chapter 10).

***Read this first.** These are starting points for a typical pop/rock context, not laws. Genre, arrangement and the specific recording change everything. Always: **subtract before you add, process in the full mix, and trust your ears over any number here.** Reverb/delay should live on **sends** (Chapter 10.5), not inserts.*

11.1 Electric guitar

Chain: Gate → EQ (cut) → Compression (light) → EQ (tone) → [Amp/NAM + IR if not already tracked] → sends: Delay, Reverb

- **Gate** (esp. high-gain): threshold just above the noise floor to silence hiss between riffs (Chapter 5 gain = noise). Fast-ish release.
- **EQ:** HPF ~80–100 Hz; cut mud ~300–500 Hz if congested; preserve the 200 Hz–1 kHz body; gentle presence boost ~3–5 kHz; tame fizz with a high-shelf cut above ~7 kHz if harsh (better: fix the IR/mic, Chapter 6).
- **Compression:** light — ratio ~2:1–4:1, a few dB reduction, medium attack to keep pick attack. Clean rhythm/funk benefits more; heavily distorted guitar is already compressed by the distortion, so go easy.
- **Effects:**
 - **Overdrive/distortion** = the amp/NAM itself (Chapters 5, 8).
 - **Delay:** tempo-synced 1/8 or dotted-1/8 for leads/ambience; low feedback (2–4 repeats), moderate level. Slapback for rockabilly.
 - **Reverb:** spring or room for vintage; plate/hall for leads. Keep rhythm parts fairly dry; feature leads more.
 - **Modulation:** chorus/phaser on clean parts for shimmer; flanger for special moments.
- **Doubling:** record two takes, pan hard L/R for a wide, powerful rhythm wall (more effective than chorus-faking width).

11.2 Acoustic guitar

Chain: EQ (cut) → Compression (gentle) → EQ (sparkle) → sends: Reverb (+ light Delay)

- **EQ:** HPF ~70–100 Hz; cut boom ~150–300 Hz; cut boxiness ~500 Hz–1 kHz if “honky”; gentle air shelf ~8–12 kHz for shimmer; presence ~2–5 kHz for pick detail if it needs to cut.

- **Compression:** gentle, ~2:1–3:1, slow-ish attack to keep the percussive pluck, to even out strumming dynamics.
- **Effects:** light plate or room **reverb** for air (high-pass the return!); a subtle 1/8 **delay** can add space to fingerpicked parts. A touch of **chorus** for a lush singer-songwriter feel. As a solo/lead instrument it can take more reverb; in a dense mix keep it drier and let it occupy ~2–5 kHz and the air band.

11.3 Bass

Chain:

EQ (clean up) → Compression (firm) → [Saturation] → [Parallel distortion] → EQ (definition)

- **EQ:** HPF very low (~30–40 Hz) to remove sub-rumble without losing weight; the power is ~80–200 Hz; cut boom ~180–250 Hz if flubby; boost **definition/growl** ~700 Hz–2.5 kHz so it's audible on phones/laptops. Carve around the **kick** (decide who owns the deep lows).
- **Compression: firmer than most instruments** — ratio ~4:1, a few dB+ reduction, medium attack/medium-fast release — to keep a steady, even foundation. Bass benefits from consistency more than dynamics.
- **Saturation:** gentle tube/tape saturation adds harmonics that make bass **translate on small speakers** (you “hear” the bass even where the lows can't reproduce — same principle as the definition boost). Very effective.
- **Parallel distortion** (a distorted send blended under the clean) is a classic rock/metal trick for grind and presence while keeping the clean low end.
- **Generally avoid** reverb/delay/heavy modulation on bass — they smear the low end and muddy the mix. Keep bass tight, focused, mono in the lows.

11.4 Keyboard / piano / synth

Chain: EQ → Compression (to taste) → [Modulation] → sends: Reverb, Delay

- **EQ:** focus on **50–250 Hz** for warmth vs. mud; HPF bright pads/organs/leads so they don't crowd bass; cut low-mid mud ~200–500 Hz; presence ~3–6 kHz for a piano/lead to come forward; air shelf ~10–14 kHz for polish. Wide instruments mask vocals — **carve a dip where the vocal sits**.
- **Compression:** acoustic/electric piano benefits from gentle leveling (~2:1–3:1); pads usually need little; plucky synths can take more for consistency.
- **Effects:**
 - **Reverb:** hall/plate suits piano and pads for lushness; keep leads tighter.
 - **Delay:** great on electric piano (Rhodes) and synth leads/arps; tempo- synced.
 - **Modulation:** **chorus** is iconic on Rhodes/electric piano and synth pads (width/shimmer); **phaser** on Rhodes/clavinet for funk; tremolo on electric piano/organ.
- **Stereo:** synths/pads can be spread wide; keep deep-bass synths mono-centered.

11.5 Vocals (lead)

Chain: Gate/cleanup → EQ (subtractive) → Compression → De-esser → EQ (tone/air) → [Saturation] → sends: Delay, Reverb

- **EQ (subtractive): HPF ~80–120 Hz** (higher for thin voices); cut chest/mud ~200–400 Hz; cut boxiness ~500 Hz–1 kHz if needed.
- **Compression:** vocals have huge dynamic range, so compress **firmly, often in two stages**: a first compressor ~3:1–4:1 catching peaks, optionally a second gentler one for consistency. Aim for an even, upfront vocal where every word is audible. Medium attack (keep consonant clarity), auto/medium release.
- **De-esser:** tame sibilance ~5–9 kHz (Chapter 10) — *after* compression (which can exaggerate “s”). Don’t just EQ-cut it (dulls the voice).
- **EQ (additive):** presence/intelligibility boost ~3–6 kHz; **air shelf ~10–16 kHz** for the “expensive” sheen; small warmth bump ~100–200 Hz if thin (watch mud).
- **Saturation:** a little adds density and helps the vocal cut without raw volume.
- **Effects (on sends):**
 - **Delay:** the main width/depth tool for vocals — keeps clarity better than heavy reverb. Tempo-synced 1/4 or 1/8, modest feedback. A **slapback** (~80–120 ms) thickens. **Haas** (<40 ms L/R) widens. Pro trick: a delay that only blooms in the *gaps* between phrases.
 - **Reverb:** plate is the classic vocal reverb. Use **pre-delay (~20–60 ms)** so the dry vocal stays in front and intelligible; **high-pass and low-pass the reverb return** so it doesn’t muddy; consider **ducking** the reverb under the vocal (compress the return, keyed by the vocal) so it swells in the gaps — keeps lyrics clear while sounding lush.
 - **Combine** a little reverb with delay rather than drowning in reverb alone.

11.6 Backing vocals

- EQ similar to lead but you can **carve more** (cut some presence ~3–5 kHz) so they sit *behind* the lead and don’t compete.
- **More effects than the lead:** more reverb, more delay, heavier **chorus** for width, harder compression for a smooth blended bed.
- **Double/triple and pan wide** for a big, lush stack; the lead stays centered and drier so it reads as the focus.

11.7 A universal reminder

Effects are seasoning. The order of priority is always: **good source** → **good arrangement** → **levels/balance** → **panning** → **EQ** → **compression** → **then time/space effects**. If something sounds wrong, fix it earlier in that list before adding another plugin. The best mixers use **fewer, more intentional** moves — not more plugins.

Summary

- **Electric guitar:** gate (if high-gain) → clean-up EQ → light comp → amp/NAM
 - IR → modest delay/reverb on leads; double-track for width.
 - **Acoustic:** HPF + de-boom/de-box EQ → gentle comp → air → light reverb.
 - **Bass:** HPF low → firm compression → saturation for translation → definition boost ~700 Hz–2.5 kHz; **avoid time effects**, keep lows mono.
 - **Keys/synth:** manage 50–250 Hz warmth/mud, carve for the vocal; chorus on Rhodes/pads; reverb/delay to taste.
 - **Vocals:** HPF → subtractive EQ → firm (often two-stage) compression → de-esser → presence/air EQ → delay (primary) + EQ'd, pre-delayed, ducked reverb on sends. Backing vocals: more effects, panned wide, sitting behind.
 - Effects are **seasoning** — fix problems earlier in the chain first.
-

***Try this.** Build a vocal chain in this exact order on one take and bypass each stage in turn: hear the HPF clear the rumble, the compression even the words, the de-esser tame the “s,” the air shelf add sheen, the delay add space. Then disable them all and re-enable only the three that helped most. That “less is more” pass is how pro mixes stay clean.*

Chapter 12 — Getting Started with REAPER

Everything in this book happens *somewhere* — and for most home studios that somewhere is a **DAW** (Digital Audio Workstation). This chapter introduces **REAPER**, the lightweight, deeply customizable, and famously affordable DAW that's become a favorite of home recordists and pros alike. We'll cover why it's a great choice, the interface, and the core workflow: tracks, recording, FX, routing/buses, MIDI, and a first mix.

12.1 Why REAPER

- **It's light and fast.** Tiny download, runs well on modest computers, starts instantly.
- **It's cheap and honest.** A fully-functional, time-unlimited evaluation, and a very affordable license (a discounted personal tier vs. a commercial tier) with no subscription — unusually fair pricing.
- **It's endlessly customizable.** Layouts, themes, actions, and scripts can be reshaped to fit how you work; you can even save multiple layouts for different tasks (tracking vs. mixing).
- **It supports everything.** All common plugin formats (VST/VST3, AU on Mac, etc.) and audio interfaces — so the free **NAM** plugin and your **IR loader** (Chapters 7–8) drop straight in.

The trade-off: REAPER ships fairly bare and expects you to set it up. That flexibility is the point, but it means a little initial configuration.

12.2 First-time setup: tell REAPER about your hardware

Before anything, point REAPER at your audio interface:

1. **Options** → **Preferences** → **Audio** → **Device**.
2. Choose the right **audio system** (on Windows, prefer **ASIO** with your interface's driver for low latency; on Mac, Core Audio).
3. Select your interface as input and output device.
4. Set a **buffer size**: *smaller* (e.g., 128 samples) = lower latency for recording but more CPU; *larger* (e.g., 512–1024) = more stable for mixing. Many people track small and mix large.
5. Set the project **sample rate / bit depth** for recording in **Project Settings** (recall Chapter 2: 44.1 or 48 kHz, 24-bit).

12.3 The interface at a glance

The main REAPER window has a few key zones:

- **Transport bar** (bottom or top): the **play**, **stop**, **record** buttons, plus the playhead position, tempo, time signature and loop controls.
- **Track Control Panel (TCP)** — the left side of each track row. This is where you manage each track: its **name**, **volume**, **pan**, **mute (M)**, **solo (S)**, **record-arm (the red button)**, **input selector**, and **FX button**.

- **Arrange / timeline area** — the large central area where recorded audio and MIDI appear as **items** (clips) on a timeline you arrange and edit.
- **Mixer (MCP)** — press **Ctrl/Cmd+M** to toggle a traditional vertical mixer view (faders, meters, FX, sends) — handy when mixing.
- **Master track** — the final output bus everything sums into before your speakers.

REAPER is heavily reconfigurable, so layouts vary — but those elements exist in every setup.

12.4 Tracks

A **track** holds audio or MIDI and its processing. To work with them:

- **Add a track:** double-click empty space in the TCP, or press **Ctrl/Cmd+T**, or right-click → *Insert new track*.
- **Name it** (double-click the name) — stay organized from the start.
- Each track has its **volume, pan, mute, solo** right there in the TCP.
- Drag to **reorder**; use track **folders** to group related tracks (e.g., all drums, all guitars) — drag a track slightly under another to nest it.

12.5 Recording audio (e.g., a guitar or vocal)

1. **Create a track** and name it.
2. **Set the input.** Click the input dropdown on the track (the little input/monitor area in the TCP) and choose the interface input your mic/guitar is plugged into (mono input 1, for example).
3. **Record-arm** the track: click the round **red arm button** in the TCP. Its meter goes live — check you're getting signal with **headroom** (peaks around **-12 to -18 dBFS**, Chapter 2).
4. **Enable input monitoring** (the speaker/monitor icon) if you want to hear yourself through the DAW (and through any amp sim) while recording.
5. **Press Record** (the transport record button, or **R**). Play/sing.
6. **Press Stop** (or **Spacebar**). A new audio **item** appears on the track.
7. Records to your chosen project format (24-bit WAV, Chapter 3) in the project folder. Use takes/loop-recording to comp multiple passes.

***Latency tip.** If monitoring through the DAW feels laggy, lower the buffer size (12.2). If your interface has direct/hardware monitoring, use that for zero-latency tracking and disable software input monitoring.*

12.6 FX: putting plugins on a track

Everything from Chapters 5–11 lives in the **FX chain**:

1. Click the **FX** button on the track (in the TCP or MCP).

2. The FX browser opens — **double-click** a plugin (EQ, ReaComp, NAM, your IR loader, reverb...) to add it.
3. Plugins run **top to bottom** in the chain window — drag to reorder (remember the order logic from Chapter 10.6: gate → EQ → comp → ... and amp/NAM **before** IR).
4. REAPER includes a full suite of stock plugins (the “**ReaPlugs**”: **ReaEQ**, **ReaComp**, **ReaGate**, **ReaVerb** [a convolution reverb that loads IRs!], **ReaDelay**, etc.) — enough to mix an entire record before buying anything.

To build a guitar tone in the box: on a track with a recorded DI, add **NAM** → then **ReaVerb** or a **dedicated IR loader** with a cab IR → then EQ/effects.

12.7 Routing and buses (sends)

This is where REAPER shines and where Chapter 10.5’s send concept becomes real.

Buses (submix groups)

A **bus** is just a track that other tracks feed into, so you can process them together (e.g., a “Drum Bus” or “Guitar Bus” with one compressor gluing the group). In REAPER, the easiest way is **track folders**: put all your guitars in a folder, and the **folder’s parent track acts as a bus** — add an EQ/compressor there to process all guitars at once.

Sends and receives (for reverb/delay)

To set up a shared reverb (Chapter 10.5):

1. Create a new track, name it “Reverb,” and put a reverb plugin on it set to **100% wet**.
2. On a vocal/guitar track, click the **Routing button** (the “**I/O**” / route button in the TCP) and add a **Send** to the “Reverb” track.
3. Adjust the **send level** to taste. Do the same from other tracks — now they **share one reverb space**, and you can EQ/compress the Reverb track’s return independently.

Everything ultimately routes to the **Master track**, which feeds your interface outputs.

12.8 MIDI and virtual instruments (keys/synths)

For keyboard, synth and programmed parts (Chapter 11.4):

1. Insert a track and add a **virtual instrument** (a VSTi synth/piano) as the first item in its **FX chain** (or use *Insert → Virtual instrument on new track*).
2. **Record-arm** the track and set its input to your **MIDI keyboard**.
3. Record-enable and play — REAPER records **MIDI items** (note data, not audio), which you can edit freely in the **piano-roll MIDI editor** (double-click a MIDI item).
4. Because it’s MIDI, you can change notes, timing, velocity, *and the instrument/sound itself* after the fact — total flexibility.

12.9 A first-mix workflow

Pulling the whole book together, a sane order of operations:

1. **Organize:** name and color tracks; folder them into groups (drums, guitars, vocals).
2. **Gain stage / balance:** rough fader levels so everything's audible with headroom on the master (no clipping — watch the master meter).
3. **Pan:** place instruments across the stereo field (lead vocal/bass/kick center; guitars and overheads out wide).
4. **EQ (subtractive):** HPF everything that isn't bass/kick; carve mud and clashes (Chapter 9).
5. **Compression:** control dynamics where needed (Chapter 10).
6. **Buses:** group with folders; glue groups with bus compression/EQ.
7. **Time/space effects on sends:** shared reverb and delay (Chapters 10–11).
8. **Automation:** ride levels and effect amounts over the song (volume envelopes, etc.).
9. **Master bus:** gentle bus processing and a **limiter**, mastering to a sensible **LUFS** target (Chapter 3), not maximum loudness.
10. **Export / Render:** *File* → *Render*. Render a **24-bit WAV master** (Chapter 3), and additionally an MP3/AAC for sharing — with **dither** when rendering to 16-bit.

Summary

- **REAPER** is light, affordable, customizable, and supports all plugins (including free **NAM** and **IR** loaders) — a great home-studio DAW that needs a little setup.
- Configure your **audio device/driver and buffer size** first; set project **44.1/48 kHz, 24-bit**.
- Core layout: **transport, Track Control Panel (arm/input/vol/pan/mute/solo/FX), arrange timeline, mixer (`Ctrl/Cmd+M`), master track**.
- **Record:** name track → set input → red-arm → check headroom (-12 to -18 dBFS) → monitor → **R** to record, **Space** to stop.
- **FX:** add plugins to a track's FX chain (top-to-bottom order); the stock **ReaPlugs** can mix a whole record; build tone with **NAM** → **IR**.
- **Routing: folders = buses** for group processing; **sends** to an FX track = shared reverb/delay; everything sums to the **Master**.
- **MIDI** tracks drive virtual instruments and are fully editable after recording.
- Mix in a sane **order** and **render** a 24-bit WAV master (plus lossy copies, with dither when down-converting).

Try this. Make a tiny project: one MIDI track with a piano VSTi, one recorded guitar through NAM → an IR, and a shared reverb on a send that both feed. Balance, pan, add a touch of EQ, and render a WAV. You've just used every major concept in this book — digital audio, the signal chain, IR/NAM, EQ, effects, routing and export — in one pass.

Appendix A — Instrument Frequency Chart

A consolidated quick-reference for EQ (Chapter 9) and effects (Chapters 10–11). **Every value is a starting point** — sweep around it and judge in the full mix.

The spectrum at a glance

Zone	Range	Character	Typical move
Sub-bass	20–60 Hz	Felt, not heard; rumble	HPF most non-bass tracks
Bass	60–250 Hz	Weight, power, fundamentals	Power <i>and</i> mud both live here
Low-mids	250–500 Hz	Warmth / mud / boxiness	Most common cut zone
Mids	500 Hz–2 kHz	Body, “honk,” attack	Cut box ~400–600 Hz; nasal ~1 kHz
High-mids	2–6 kHz	Presence, clarity, bite & harshness	Boost to cut through; cut if harsh
Highs	6–12 kHz	Brilliance, detail, sibilance	Sibilance ~5–9 kHz (de-ess)
Air	12–20 kHz	Openness, sheen	Gentle high shelf for sparkle

Per-instrument cheat sheet

HPF = high-pass filter (low-cut). All values approximate.

Electric guitar

Goal	Frequency
HPF (remove low rumble)	80–100 Hz
Body / “meat” (preserve)	200 Hz–1 kHz
Mud / congestion (cut)	300–500 Hz
Presence / bite (boost)	2.5–5 kHz
Fizz (high-cut/shelf if harsh)	above 6–8 kHz

Acoustic guitar

Goal	Frequency
HPF	70–100 Hz
Boom (cut)	150–300 Hz
Boxiness / honk (cut)	500 Hz–1 kHz

Goal	Frequency
Pick attack / clarity (boost)	2–5 kHz
Sparkle / air (boost)	8–12 kHz

Bass guitar

Goal	Frequency
HPF (sub-rumble only)	30–40 Hz
Bottom / sub	40–80 Hz
Fundamentals / power (“the magic”)	80–200 Hz
Boom (cut if flubby)	180–250 Hz
Definition / growl (boost — translates on small speakers)	700 Hz–2.5 kHz
String/pick clack	2.5–5 kHz

Keyboard / piano / synth

Goal	Frequency
HPF (bright pads/organs/leads)	60–120 Hz
Warmth vs. mud (focus zone)	50–250 Hz
Mud / boxiness (cut)	200–500 Hz
Presence / clarity (boost)	3–6 kHz
Air / polish (high shelf)	10–14 kHz

Vocals

Goal	Frequency
HPF (min; higher for thin voices)	80–120 Hz
Warmth / body	100–250 Hz
Mud / chestiness (cut)	200–400 Hz
Boxiness / honk (cut)	500 Hz–1 kHz
Presence / intelligibility (boost)	3–6 kHz (clarity often 4–9 kHz)
Sibilance (de-ess, don’t EQ)	5–9 kHz
Air / sheen (high shelf)	10–16 kHz

Effect quick-reference

Effect	Use it for	Key control
Compression	Even out dynamics, add punch/glue	Threshold, ratio, attack, release
Limiter	Cap peaks, master loudness	Ceiling
Gate	Silence noise/bleed (high-gain gtr)	Threshold
De-esser	Tame “s”/“sh” on vocals	Frequency (~5–9 kHz)
Reverb	Depth, space, ambience	Type, decay, pre-delay (EQ the return!)
Delay	Echo, width, clarity-friendly space	Time (tempo-sync), feedback
Chorus	Width/shimmer (clean gtr, Rhodes, pads, BVs)	Depth, rate
Saturation	Warmth, density, small-speaker translation	Drive

Reverb/delay rules of thumb

- Put reverb/delay on **sends**, not inserts (share the space; EQ the return).
- **High-pass** reverb returns ~200–300 Hz; roll off highs to sit it back.
- **Pre-delay** ~20–60 ms keeps the dry source in front.
- **Haas** (<40 ms L/R) = width without an audible echo.
- **Slapback** ~80–120 ms thickens vocals/rockabilly guitar.

Appendix B — Glossary

Quick definitions of the key terms used in this book. Chapter references point to where each is explained in depth.

A/D converter (ADC) — Analog-to-digital converter; turns the analog voltage from a mic/instrument into digital samples. (Ch. 2)

AAC — Advanced Audio Coding; a lossy format with better quality than MP3 at the same bitrate; standard on Apple Music/YouTube. (Ch. 3)

Aliasing — False, lower frequencies created when a signal contains content above the Nyquist frequency; prevented by an anti-aliasing filter. (Ch. 2)

Amplitude — The size of a sound wave's pressure swing; perceived as loudness. (Ch. 1)

Bit depth — How precisely each sample's amplitude is measured; determines dynamic range (~6 dB per bit). 16-bit ≈ 96 dB, 24-bit ≈ 144 dB. (Ch. 2)

Bitrate — Data per second (kbps) in a lossy file; higher = better quality & bigger files. (Ch. 3)

Bus — A track other tracks feed into, for processing a group together (e.g., a drum bus). In REAPER, often a track folder's parent. (Ch. 10, 12)

Cabinet (cab) — The speaker + enclosure that filters and colors an amp's output; a strong, characterful, **linear** filter. (Ch. 6)

Clipping — Flattening a wave's peaks past a system's limit; mathematically identical to adding harmonics (= distortion). Digital clipping above 0 dBFS is harsh/broken. (Ch. 2, 5)

Compression (data) — Making a file smaller; lossless (reversible) or lossy (discards data). (Ch. 3)

Compression (dynamics) — An effect that reduces the level difference between loud and quiet parts. (Ch. 10)

Convolution — The math operation that applies an IR to a signal, “stamping” the captured system's character onto it. (Ch. 7)

dBFS — Decibels relative to Full Scale; digital level where **0 = the ceiling** and values count down (negative). (Ch. 2)

Decibel (dB) — A logarithmic ratio unit for level/loudness. (Ch. 1)

De-esser — A frequency-targeted compressor that ducks sibilance (~5–9 kHz). (Ch. 10)

Delay — An effect that repeats the signal after a set time; adds echo/width. (Ch. 10)

Dither — Tiny shaped noise added when reducing bit depth, to avoid quantization distortion; applied **once** at final export. (Ch. 2)

Dynamic range — The span between the quietest and loudest a system handles; set by bit depth in digital. (Ch. 2)

Envelope — How a sound's loudness evolves over time: attack, decay, sustain, release. (Ch. 1)

EQ (equalization) — Boosting/cutting bands of frequencies; a **linear** process. (Ch. 9)

FLAC — Free Lossless Audio Codec; lossless compression to ~half of WAV size. (Ch. 3)

Frequency — Cycles per second (Hz); perceived as pitch. Human hearing $\approx$ 20 Hz–20 kHz. (Ch. 1)

Fundamental — The lowest frequency of a note, which sets its perceived pitch. (Ch. 1)

Gain staging — Setting levels through a chain; in amps, cascading multiple gain stages to build controllable distortion. (Ch. 5)

Gate (noise gate) — Silences/attenuates signal below a threshold; kills high-gain noise and bleed. (Ch. 10)

Haas effect — A copy delayed under $\sim$ 40 ms is heard as width/thickness, not a separate echo. (Ch. 10)

Harmonics (overtones) — Frequencies above the fundamental (near-integer multiples); their balance is **timbre**. Distortion *adds* harmonics. (Ch. 1, 5)

Headroom — The gap between your signal's peaks and 0 dBFS; record with $\sim$ 12–18 dB of it. (Ch. 2)

High-pass filter (HPF) / low-cut — Removes frequencies below a chosen point; the most-used mixing move. (Ch. 9)

Impulse Response (IR) — A recording of a **linear** system's response to an impulse; an acoustic fingerprint of a cab+mic+room. Applied via convolution. (Ch. 7)

Insert — An effect placed directly in a channel (whole signal passes through); vs. a send. (Ch. 10)

Latency — Delay through the recording system; reduced with smaller buffer sizes. (Ch. 12)

Limiter — A compressor with very high ratio/fast attack that caps peaks at a ceiling. (Ch. 10)

Linear / Nonlinear — Linear = changes level/frequency balance, adds no new frequencies (EQ, cabinets). Nonlinear = creates new harmonics (distortion, preamps). Determines IR vs. NAM. (Ch. 5, 7)

Lossless / Lossy — Lossless keeps all audio data (WAV/FLAC); lossy discards some permanently (MP3/AAC/Opus). (Ch. 3)

Low-pass filter (LPF) / high-cut — Removes frequencies above a chosen point; tames fizz/harshness. (Ch. 9)

LUFS — Loudness Units Full Scale; the loudness measure streaming platforms normalize to (~ -14 LUFS). (Ch. 3)

MIDI — Note/performance data (not audio) that drives virtual instruments; fully editable after recording. (Ch. 12)

Modulation effects — Chorus, flanger, phaser, tremolo, vibrato; use an LFO for movement/width. (Ch. 10)

MP3 — The original mass-market lossy format; universally compatible. (Ch. 3)

NAM (Neural Amp Modeler) — Free, open-source tool that captures **nonlinear** amps/pedals/chains by training a neural network on input $\rightarrow$ output audio. (Ch. 8)

Nyquist frequency — Half the sample rate; the highest frequency a given rate can capture. (Ch. 2)

Nyquist–Shannon theorem — You must sample at $>2\times$ the highest frequency to capture it; basis of the 44.1 kHz standard. (Ch. 2)

Opus — Modern open lossy codec; best quality at low/mid bitrates; common for voice/streaming. (Ch. 3)

PCM (Pulse Code Modulation) — The basic scheme of storing audio as a stream of sampled amplitude values; underlies WAV/AIFF/CDs. (Ch. 2, 3)

Phase — The timing alignment of waves; aligned waves reinforce, opposed waves cancel. (Ch. 1)

Preamp — First amplification stage; boosts and tone-shapes, and (driven) adds the amp's distortion character. (Ch. 5)

Pre-delay — The gap before a reverb's tail starts; keeps the dry source clear. (Ch. 10)

Profiling — Capturing a real amp's sound by analysis (NAM; commercially, Kemper). (Ch. 8)

Q — A bell filter's bandwidth; low Q = wide, high Q = narrow. (Ch. 9)

Quantization error — Rounding error from limited bit depth; heard as distortion on quiet passages; mitigated by dither. (Ch. 2)

Reverb — Simulates a space's reflections; adds depth/ambience. (Ch. 10)

Sample / Sample rate — A single amplitude measurement / how many per second (Hz). 44.1 kHz = 44,100/sec. (Ch. 2)

Saturation — Gentle harmonic distortion for warmth/density/translation. (Ch. 10)

Send (aux) — A copy of a channel routed to a separate FX track (e.g., shared reverb). (Ch. 10, 12)

Shelf filter — Boosts/cuts everything above (high shelf) or below (low shelf) a frequency. (Ch. 9)

Sibilance — Harsh "s"/"sh" energy ($\sim 5\text{--}9$ kHz); tamed with a de-esser. (Ch. 9, 10)

Sine wave — A single pure frequency; the building block of all sound; sounds “electronic.” (Ch. 1)

Slapback — A single short echo (~80–120 ms) for thickening. (Ch. 10)

Timbre — Tone color; determined by the harmonic recipe and envelope. (Ch. 1)

Threshold — The level at which a dynamics processor (comp/gate) starts acting. (Ch. 10)

WAV / AIFF — Uncompressed PCM containers; the working/master format. (Ch. 3)

Appendix C — Sources & Further Learning

This book synthesizes established audio-engineering knowledge with current, web-researched references. Below are the primary sources consulted while writing, grouped by topic, followed by suggestions for going deeper.

Sources consulted

Digital audio: sample rate, bit depth, Nyquist (Chapter 2)

- [iZotope — Digital audio basics: sample rate and bit depth](#)
- [Sonarworks — Understanding Sample Rate](#)
- [Skytracks — Sample Rate & Bit Depth Explained \(44.1 vs 48 vs 96 kHz\)](#)
- [MajorMixing — Audio Sample Rate and Bit Depth: Complete Guide](#)

Audio formats & compression (Chapter 3)

- [What Hi-Fi? — MP3, AAC, WAV, FLAC: all the audio file formats explained](#)
- [LANDR — Audio File Formats: AIFF, WAV, MP3, Ogg, FLAC](#)
- [Feedtracks — Audio File Formats Explained: WAV, FLAC, MP3, AAC, OGG \(2025\)](#)
- [LALAL.AI — Differences Between Audio Formats](#)

Preamps, gain stages & harmonic distortion (Chapter 5)

- [Effectrode — Guitar Preamp Tone Explained](#)
- [Premier Guitar — Everything You Ever Wanted to Know About Preamp Tubes](#)
- [Tonestack.net — Tube vs. Solid State: Is Tube Distortion a Myth?](#)
- [Carvin Audio — Preamp and Power Amp Tubes: What They Are and What They Do](#)
- [Home Studio Guys — Understanding Guitar Preamps and Their Role in Tone](#)

Cabinets, mics & impulse responses (Chapters 6–7)

- [TONE3000 — Impulse Response \(IR\) guide](#)
- [TONE3000 — What Is an Impulse Response \(IR\)? Beginner's Guide for Guitarists](#)
- [Premier Guitar — The Working Guitarist: All About Impulse Responses](#)
- [Sweetwater InSync — How to Use Speaker Impulse Responses with Guitar Amplifiers](#)
- [Chaos Audio — What Are Impulse Responses \(IRs\) and Why Do They Matter?](#)

Neural Amp Modeler (Chapter 8)

- [Neural Amp Modeler — official site](#)
- [TONE3000 — Neural Amp Modeler guide](#)
- [TONE3000 — Capture your gear \(profiling\)](#)
- [Nail The Mix — A Guide to Neural Amp Modeler \(NAM\)](#)

- [gearnews.com — Neural Amp Modeler: Plugin, Profiles, Hardware — Complete Guide](#)
- [Chernobyl Audio — Neural Amp Modeler \(NAM Profiles\) Explained](#)

EQ & frequency charts (Chapter 9, Appendix A)

- [Sonicbids — The Ultimate EQ Cheat Sheet for Every Common Instrument](#)
- [Hyperbits — The Ultimate EQ Cheat Sheet for 30+ Instruments](#)
- [Sweetwater InSync — EQ Frequencies of Musical Instruments Explained](#)
- [Audio Issues — The Only EQ Chart You’ll Ever Need](#)
- [Produce Like A Pro — EQ Cheat Sheet](#)

Effects: compression, reverb, delay, modulation (Chapters 10–11)

- [Waves — 7 Ways to Enhance Reverb & Delay with Compression & EQ](#)
- [Waves — 4 Famous Vocal Mixing Tricks Using Reverb and Delay](#)
- [Music Guy Mixing — Chorus, Reverb, Delay: How They Differ and Which to Use](#)
- [Reverb.com — How to Use Reverb and Delay Tastefully on Vocals](#)

REAPER (Chapter 12)

- [LANDR — Reaper DAW: The Ultimate Tutorial for Producers](#)
- [Mastering.com — Reaper Tutorial: Up and Running in 20 Minutes](#)
- [Unison — Reaper DAW 101: The Official Beginner’s Guide](#)
- [Pro Mix Academy — How To Record In REAPER \(Step-by-Step Guide\)](#)
- [MakeUseOf — How to Get Started With REAPER: A Beginner’s Guide](#)

Where to go next

- **Official REAPER resources.** The REAPER User Guide (the “ReaperMania” community videos and Kenny Gioia’s official tutorial series on YouTube) are the fastest way to get fluent in the DAW.
- **Get the free tools.** Download the **NAM** plugin from its official site and grab free captures and IRs from **TONE3000** — then experiment with the “Try this” boxes in Chapters 7–8.
- **Train your ears.** Ear-training tools (e.g., frequency-identification games) turn the EQ map of Chapter 9 into instinct faster than reading ever will.
- **Learn loudness.** Read up on **LUFS** and streaming loudness normalization (Chapter 3) before you master anything for release.
- **Study reference mixes.** Pull professionally-mixed songs in your genre into your DAW and A/B your work against them — the single most valuable habit in mixing.

A closing note. No book substitutes for hours at the faders. Use this one as a map: when you hit a term or a problem, come back, get the mental model, then go make decisions with your ears. That loop — understand, try, listen, adjust — is the whole craft.